

SCIENTIFIC JOURNAL PUBLICATION TREND TRACKING SYSTEM

Group 6

I. INTRODUCTION

1 Overview

1.1 Project Information

- Project Name: Scientific Journal Publication Trend Tracking System
- Group Name: Group 6
- Software Type: Web Application
- GitHub Repository:
<https://github.com/NguyenNgocBaoTram10/Scientific-Journal-Publication-Trend-Tracking-System>

1.2 Project team

Full Name	Role	Student ID	Email
Lê Thanh Tú	Leader	049206004778	tult4778@ut.edu.vn
Nguyễn Ngọc Bảo Trâm	Member	082306008015	tramnnb8015@ut.edu.vn
Lê Thị Huyền Trân	Member	077306003635	tranlth3635@ut.edu.vn
Trần Trung Hiếu	Member	089206008473	hieutt8473@ut.edu.vn
Trần Văn Học	Member	052206005618	hoctv5618@ut.edu.vn
Nguyễn Đoàn Nhật Quang	Member	079206010585	quangndn0585@ut.edu.vn

2 Project Scope and Limitations

2.1 Major Features

2.1.1 Major Features for User

- Register and log in to the system.
- Search scientific publications using keywords, authors, journals, or research fields.
- View detailed publication information.
- Save publications to bookmarks.
- Follow research topics of interest.

- Receive notifications about new publications and research trends.
- View publication trend analysis and statistical dashboards.
- Generate and export analytical reports (PDF/CSV).

2.1.2 Major Features for Administrator

- Log in and log out.
- Manage user accounts.
- Configure academic API sources (Semantic Scholar, OpenAlex, and Crossref).
- Monitor the data synchronization process.
- Manage system notifications.
- Generate system reports.

2.2 Project Limitations

2.2.1 Limited Use of Full-Text Data

- The system only uses publication metadata such as title, abstract, keywords, authors, and publication year.
- It does not process full-text content, which limits the depth of analysis and understanding of research papers.

2.2.2 Dependence on External APIs

- The system relies on external APIs such as OpenAlex, and Crossref.
- Any API downtime, structural changes, or request limitations may affect system stability and data collection performance.

2.2.3 Simplified Trend Analysis and Update Frequency

- Trend analysis is mainly based on publication frequency and growth rate.
- It does not apply advanced Machine Learning or AI models for prediction.
- Data is synchronized periodically (daily or weekly), so real-time trends may not be immediately reflected.

II. PROJECT MANAGEMENT PLAN

1 Overview

1.1 Scope and estimation

#	WBS Item	Complexity	Duration (Days)
1. Project Initiation			
1.1	Define project scope	Medium	1
1.2	Collect and analyze requirements	Complex	2
2. Project Management and Environment Setup			
2.1	Set up development environment (GitHub, Database, Backend, Frontend)	Simple	1
2.2	Design database (ERD, Database Schema)	Complex	2
2.3	Configure security (JWT, RBAC, Password Hashing)	Complex	2
3. System Design			
3.1	Design Use Case Diagram and Activity Diagram	Medium	1
3.2	Design Sequence Diagram	Medium	1
3.3	Design DFD and ERD	Medium	1
4. System Development			
4.1	User Registration and Login	Medium	2
4.2	Search Scientific Publications	Complex	2
4.3	View Publication Details	Medium	1
4.4	Bookmark Management	Medium	1
4.5	Research Topic Following	Medium	1
4.6	Notification Management	Medium	1
4.7	Publication Trend Analysis	Complex	3
4.8	Develop Statistical Dashboard	Complex	2
4.9	Display Trending Topics	Medium	1
4.10	Generate Statistical Reports (PDF/CSV)	Complex	2

#	WBS Item	Complexity	Duration (Days)
5. Administration Functions			
5.1	User Management	Complex	2
5.2	Configure Academic API Sources	Complex	2
5.3	Synchronize Data from Semantic Scholar, OpenAlex, and Crossref	Complex	3
6. Testing			
6.1	Prepare Test Plan	Medium	1
6.2	Design Test Cases	Medium	2
6.3	Execute Testing	Complex	2
6.4	Fix Defects After Testing	Complex	2
7. Project Finalization			
7.1	Complete Project Documentation	Medium	2
7.2	Deploy the System	Medium	1
Total Duration			45 Working Days

1.2 Project Objective

- **Timeline:** The project must be completed before **July 3, 2026**.
- **Allocated Effort:** 45 man-days.

1.3 Project Risks

#	Risk Description	Impact	Possibility	Response Plan
1	External academic APIs (Semantic Scholar, OpenAlex, Crossref) become unavailable or change their endpoints.	The system cannot retrieve publication data, causing incomplete search results and inaccurate trend analysis.	High	Implement API fallback mechanisms, monitor API status regularly, and cache previously retrieved data whenever possible.

#	Risk Description	Impact	Possibility	Response Plan
2	Large volumes of publication data lead to slow system performance.	Users experience long response times when searching publications or viewing dashboards.	Medium	Optimize database indexing, implement pagination, caching, and asynchronous background synchronization.
3	Security vulnerabilities expose user accounts or stored data.	Unauthorized access, data leakage, and loss of user trust.	Medium	Use JWT authentication, RBAC authorization, password hashing, HTTPS, and regular security testing.
4	Trend analysis results become inaccurate because of outdated or incomplete datasets.	Generated reports and statistics may mislead users' research decisions.	Medium	Schedule automatic data synchronization and validate collected data before analysis.
5	System integration between frontend, backend, and database encounters compatibility issues.	Development delays and unexpected system failures during deployment.	Medium	Perform integration testing continuously and maintain version compatibility among all components.
6	Insufficient testing before deployment leaves critical defects unresolved.	System crashes, incorrect functionality, and poor user experience.	High	Prepare comprehensive Test Plans, Test Cases, conduct functional, integration, and user acceptance testing before release.
7	Project schedule delays due to underestimated implementation effort.	Failure to complete the project within the planned timeline.	Medium	Monitor project progress weekly, prioritize critical features, and adjust the implementation schedule when necessary.

2 Project Deliverables

Sprint	Week	Duration	Deliverables	Note
Sprint 1	Week 1	14/05 – 20/05/2026	• Define project scope • Analysis requirement	
Sprint 2	Week 2	21/05 – 27/05/2026	• Finish Product Overview • Finish User Requirement • Finish Functional Requirement	
Sprint 3	Week 3	28/05 – 03/06/2026	• Finish Database Design • Finish Software design description	
Sprint 4	Week 4	04/06 – 10/06/2026	• Finish Software requirement specification	
Sprint 5	Week 5	11/06 – 17/06/2026	• Finish Software design description (continued)	
Sprint 6	Week 6	18/06 – 24/06/2026	• Finish Software testing document • Finish Release package and user guide	
Sprint 7	Week 7	25/06 – 01/07/2026	• Testing • Finish document	

3. Responsibility Assignment

Do ~D; Review ~R; Support ~S

Responsibility	Thanh Tú	Bảo Trâm	Huyền Trân	Trung Hiếu	Văn Học	Nhật Quang
Project Planning & Tracking	S	D	S	S	S	S
Prepare Project Introduction Document	R	D	R	R	R	R
Prepare Software Requirement Document	S	D	S	S	S	S
Prepare Software Design Document	D	D	D	D	D	D
Prepare Test Document	D	D	D	D	D	D
Prepare Software User Guides	D	D	D	D	D	D
Prepare Final Report	D	D	D	D	D	D

4. Project Communications

Communication Item	Who	Purpose	When	Type
Team Weekly Meeting	Group	- Report work project - Assign work for next week - Control project deadline	Every Sunday	Meeting on Google Meet
Weekly Report	Member	Report work process	1 day / week	Report via Zalo
Group Meeting	Group	Report work process	Every Thursday	Offline in class

5. Configuration Management

5.1 Document Management

- Document tool: Confluence
- Task management tool: Jira

5.2 Source Code Management

- Source code management tool: GitHub

5.3 Tools & Infrastructure

Category	Tools / Infrastructure
Programming Languages	PHP, HTML5, CSS3, JavaScript
Database	MySQL
Database Administration	phpMyAdmin
Academic APIs	OpenAlex API, Crossref REST API
API Testing	Postman
Version Control	GitHub
Project Management	Jira, Confluence
Diagramming	Draw.io, PlantUML
IDE / Code Editor	Visual Studio Code
Local Server	WAMP Server

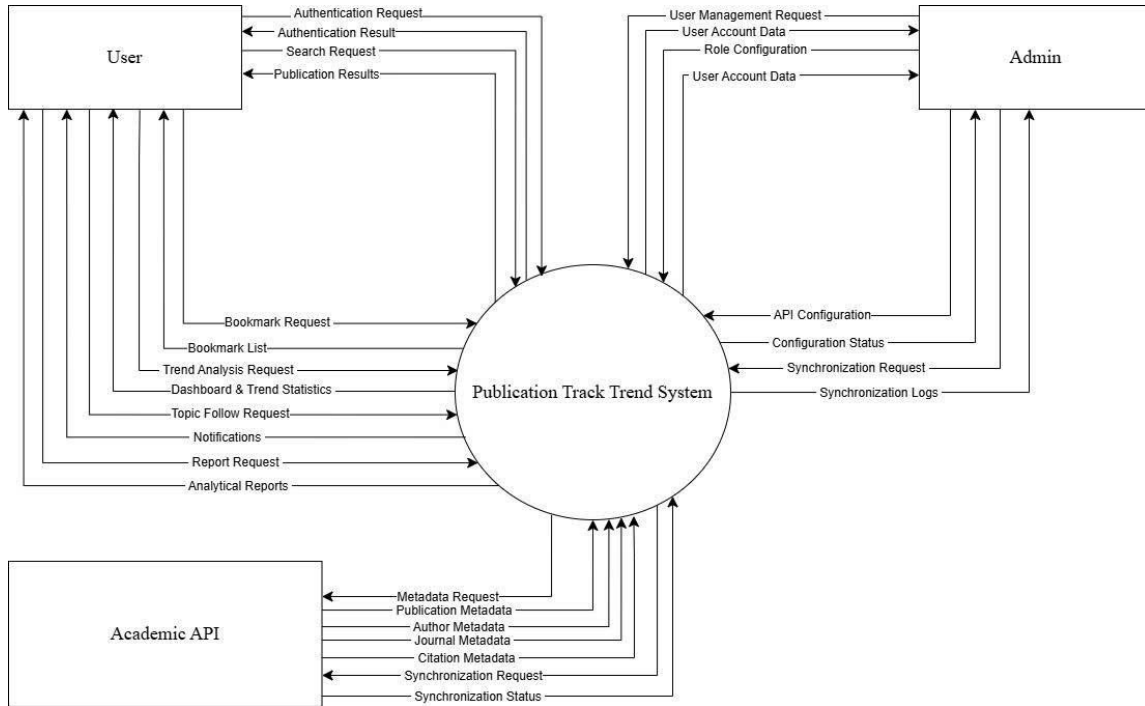
III. SOFTWARE REQUIREMENT SPECIFICATION

1 Product Overview

The Scientific Journal Publication Trend Tracking System (SJPTTS) is a web-based application developed to help researchers, students, lecturers, and academic institutions search, monitor, and analyze scientific publication trends. The system integrates academic data from OpenAlex, and Crossref to provide comprehensive and up-to-date publication information.

The system allows users to search scientific publications using keywords, authors, journals, research fields. Users can view detailed publication information, bookmark publications, follow research topics, receive notifications about newly published papers, analyze publication trends through interactive dashboards, and export reports in PDF or CSV format.

Administrators can manage user accounts, configure academic API sources, synchronize publication data, monitor system operations, and maintain the overall performance of the platform. The system aims to provide an efficient solution for discovering scientific literature and supporting research through publication trend analysis.



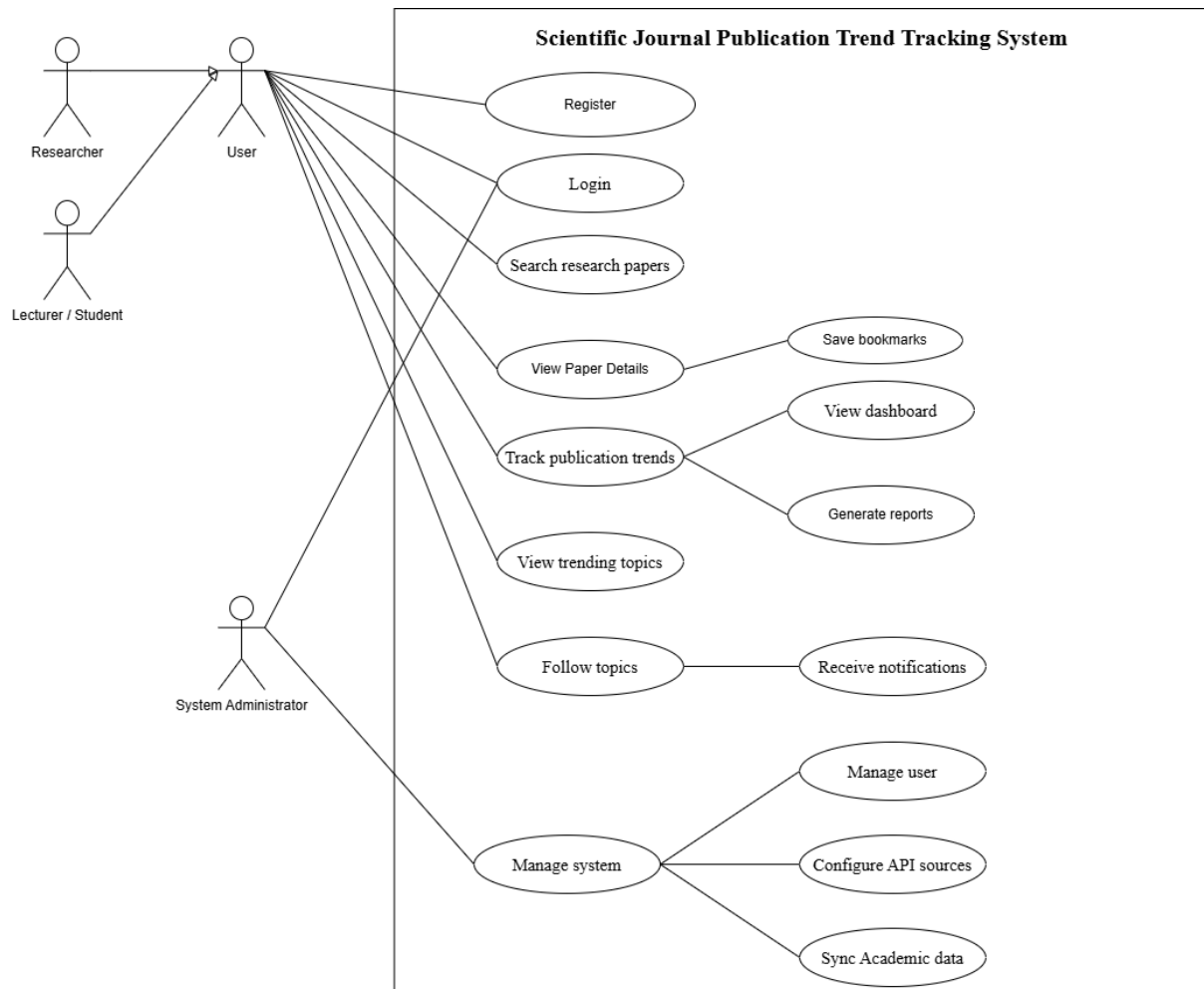
2 User Requirements

2.1 Actors

#	Actor	Description
1	User	System user who can search scientific publications, view publication details, save bookmarks, follow research topics, receive notifications, and analyze publication trends.
2	Administrator	System administrator responsible for managing users, configuring the system, setting up and validating academic data sources (OpenAlex, Crossref), synchronizing data, and monitoring system operations.
3	Publication Trend Tracking System	The core system that handles all functionalities such as paper search, trend analysis, dashboard generation, trending topic detection, bookmark management, notification delivery, and report generation.
4	Academic API	External services such as OpenAlex and Crossref that provide scientific publication metadata (authors, journals, citations) for system synchronization and analysis.

2.2 Use Case Diagrams

2.2.1 Diagram



2.2.2 Use Case Descriptions

Main UC Group	Main Agent (Actor)	Sub-UCs	Brief Description of the Process	Output
Manage Account	User	Register, Login	Users create an account and authenticate to access the Scientific Journal Publication Trend Tracking System.	A registered account is created and a secure user session is established.

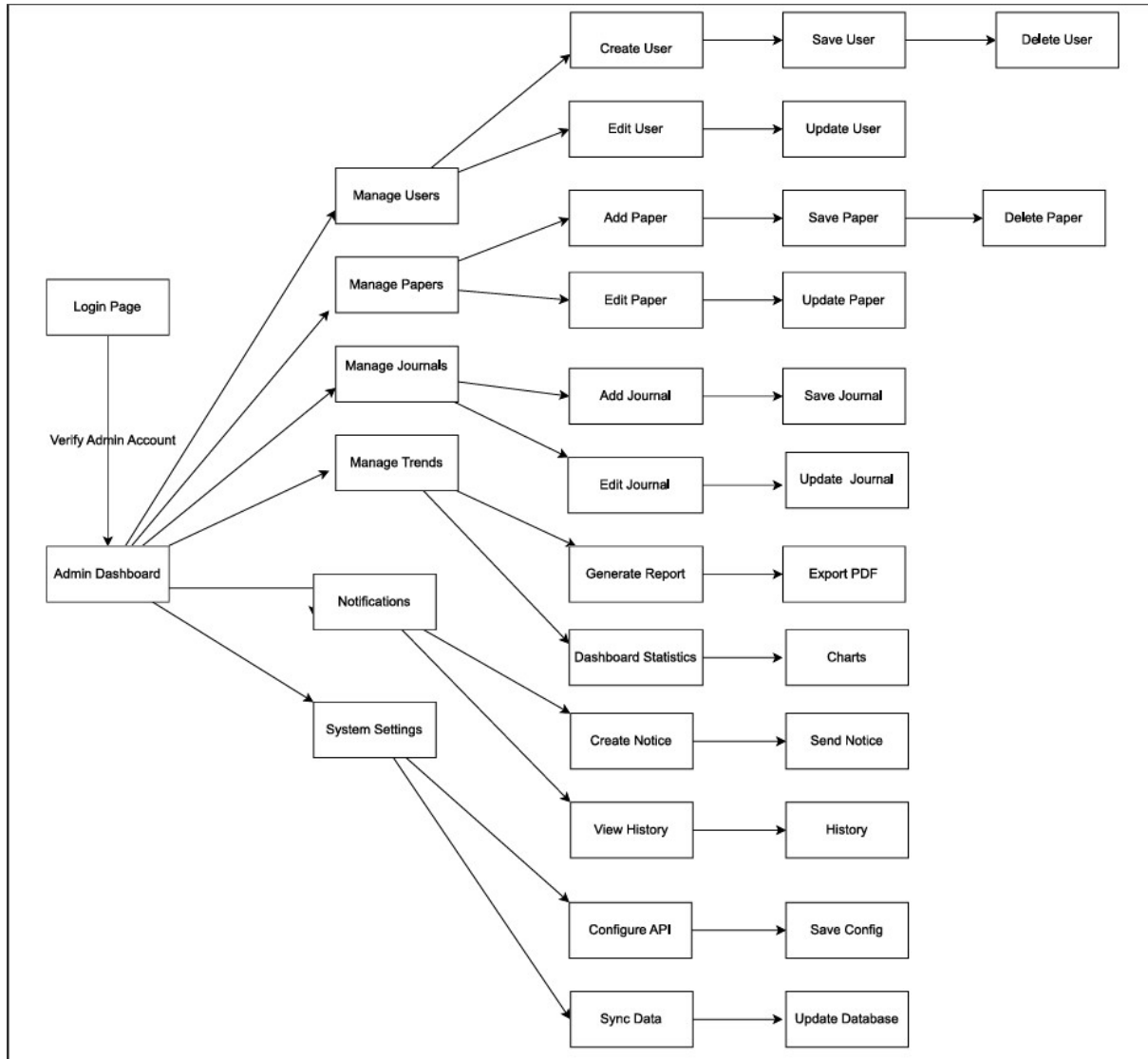
Main UC Group	Main Agent (Actor)	Sub-UCs	Brief Description of the Process	Output
Search Publications	User	Search Research Papers, View Paper Details	Users search scientific publications using keywords, author names, or journal names and view detailed publication meta-data.	Matching publications and detailed publication information are displayed.
Manage Bookmarks	User	Save Bookmarks	Users save selected publications into their personal bookmark collection for future reference.	The selected publication is stored in the user's bookmark list.
Track Publication Trends	User	Track Publication Trends, View Dashboard, View Trending Topics	Users analyze publication trends by keyword, topic, author, journal, and time range. The system generates statistical charts and identifies trending research topics.	Trend statistics, dashboards and trending topics are displayed.
Follow Topics	User	Follow Topics, Receive Notifications	Users follow research topics to receive notifications about tracked activities	The selected topic is added to the user's followed-topic list and notifications become available.
Generate Reports	User	Generate Reports	Users generate analytical reports based on publication trend analysis and selected report parameters.	An analytical report summarizing publication trends is generated.
Manage System	Administrator	Manage System, Configure API Sources, Synchronize Academic Data	Administrators manage system settings, configure academic APIs, synchronize publication metadata and maintain system operations.	System settings and publication metadata are successfully updated.
Manage Users	Administrator	Manage Users	Administrators manage user accounts, roles, permissions and account status.	User information and access permissions are updated.
Configure API Sources	Administrator	Configure API Sources	Administrators configure, update and test external academic APIs such as OpenAlex, Crossref and Semantic Scholar.	API configuration settings are successfully validated and stored.

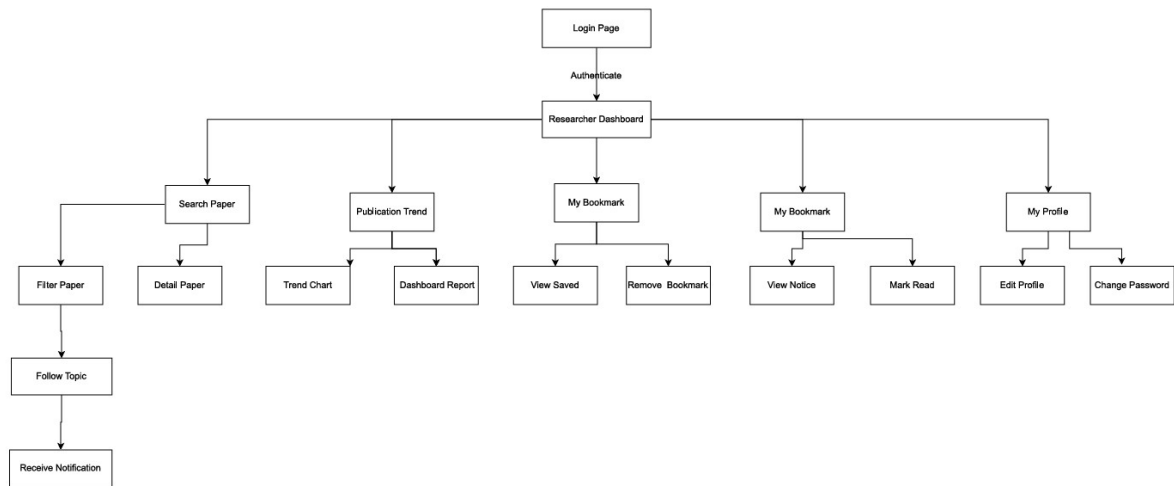
Main UC Group	Main Agent (Actor)	Sub-UCs	Brief Description of the Process	Output
Synchronize Academic Data	Administrator	Synchronize Academic Data	Administrators synchronize publication metadata from external academic APIs into the local database for searching and trend analysis.	The publication metadata database is updated with the latest synchronized records.

3 Functional Requirement

3.1 System Functional Overview

3.1.1 Screens Flow





Functional Group	Screen Name	Detailed Description
User Authentication	Login	Purpose: Allows users to log into the system. Components: Username, Password, Login button, and Register link. Flow: The user enters credentials → The system authenticates the user → The main system interface is displayed. Rule: Username and password are required; invalid credentials will generate an error message.
User Authentication	Register	Purpose: Allows new users to create an account. Components: Full Name, Email, Username, Password, and Register button. Flow: The user enters the required information → The system validates the email and username → A new account is created → The user is redirected to the Login page. Rules: Email and username must be unique; passwords are securely hashed before storage.
Paper Management	Search Research Papers	Purpose: Allows users to search scientific publications by keyword, author, or journal. Components: Search box, filters, Search button, and result list. Flow: The user enters search criteria → The system searches publication metadata → Matching publications are displayed.

Functional Group	Screen Name	Detailed Description
Paper Management	View Paper Details	Purpose: Displays detailed metadata of a selected research paper. Components: Title, Authors, Abstract, Keywords, Publication Year, Journal Name, Citation Information, and Save Bookmark button. Flow: The user selects a paper → The system retrieves publication metadata → Paper details are displayed.
Paper Management	Bookmarks	Purpose: Manages the user's bookmarked papers. Components: Bookmark list, Delete Bookmark button. Flow: The user views or removes bookmarked papers. Rule: Duplicate bookmarks are not allowed.
Publication Trend Analysis	Track Publication Trends	Purpose: Analyzes publication trends based on keywords, research topics, authors, or journals. Components: Search filters, time range selector, and Analyze button. Flow: The user enters analysis criteria → The system analyzes publication metadata → Statistical results and charts are displayed.
Publication Trend Analysis	Dashboard	Purpose: Displays an overview of publication statistics. Components: Publications by year chart, Topic distribution chart, Journal distribution chart, and Growth trend chart. Flow: The system generates dashboard visualizations after trend analysis is completed.
Publication Trend Analysis	Trending Topics	Purpose: Displays the most popular research topics. Flow: The system analyzes publication data → Ranks research topics → Displays trending topics.
Topic Management	Follow Topics	Purpose: Allows users to follow research topics of interest. Components: Topic list and Follow button. Flow: The user selects a topic → The system stores it in the followed-topic list.
Notifications	Notifications	Purpose: Displays system notifications or notifications when following a new topic. Flow: The system generates notifications → The user views notification details and marks them as read.

Functional Group	Screen Name	Detailed Description
Reporting	Generate Reports	Purpose: Generates reports based on publication trend analysis. Flow: The user clicks "Generate Report" → The system displays options: Preview & Print / Save as PDF, Direct PDF Download, and Download Report (HTML) → The system executes the request based on the selected option and completes the generation, display, or export of the analytical report.
System Administration	Manage System	Purpose: Provides administrative functions for system management. Components: User Management, API Configuration, and Academic Data Synchronization. Flow: The administrator selects a management function → The system performs the requested operation.
System Administration	Manage Users	Purpose: Manages user accounts. Components: User list, Change User Role, Activate Account, and Deactivate Account. Flow: The administrator selects a user → Updates account information → Saves the changes.
System Administration	Configure API Sources	Purpose: To define and manage external academic data sources. Flow: The administrator enters source details (Name, URL, Priority, Status, Description). The administrator clicks "Add API Source" to save it to the system. The system displays the list, allowing the administrator to track, edit, or delete entries via the "Action" column.
System Administration	Synchronize Academic Data	Purpose: Synchronize publication data from external APIs into the system. Components: API List, Synchronize button, Synchronization Log. Workflow: Click Synchronize → The system retrieves data from the APIs → Updates the database → Displays the synchronization result.

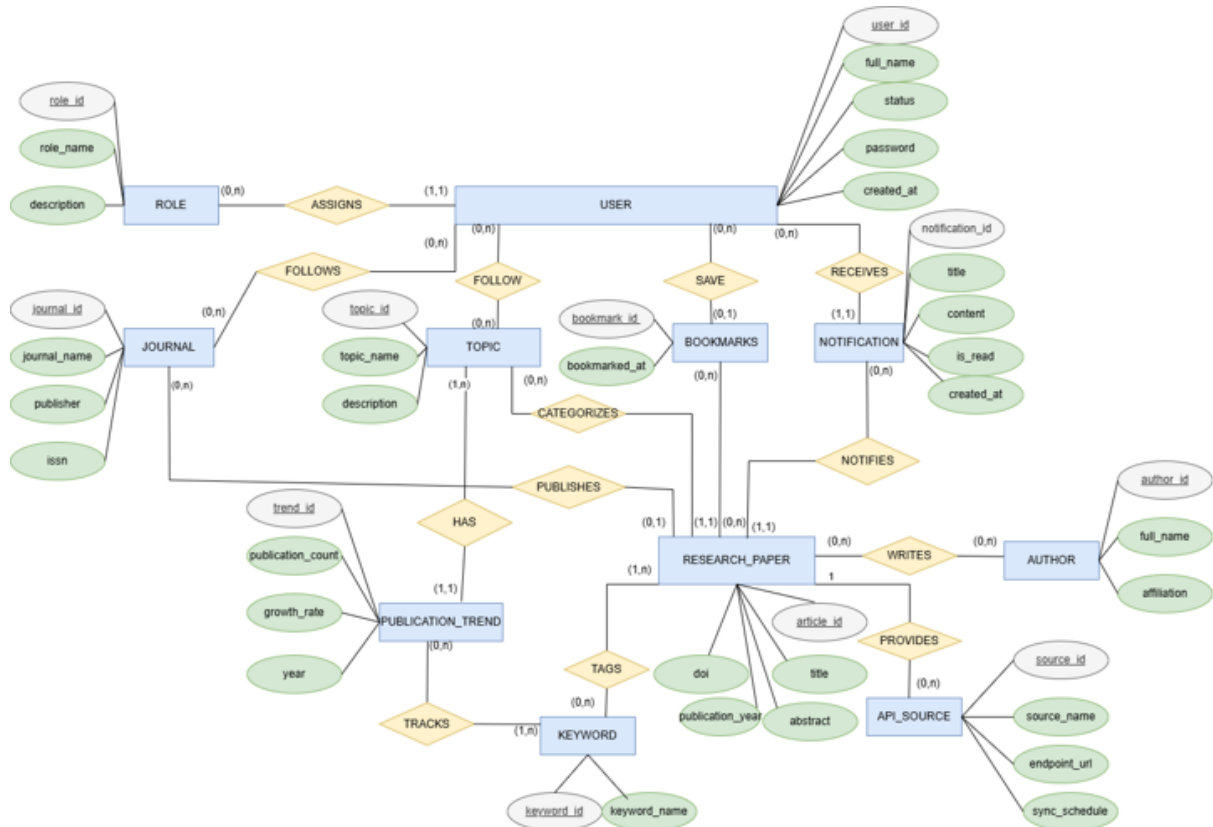
3.1.2 Screen Authorization

Screen	Researcher	Lecturer/Student	System Administrator
Login	x	x	x
Logout	x	x	x
Register	x	x	
Search Research Papers	x	x	
View Paper Details	x	x	
Save Bookmarks	x	x	
Track Publication Trends	x	x	
View Dashboard	x	x	
Generate Reports	x	x	
View Trending Topics	x	x	
Follow Topics	x	x	
Receive Notifications	x	x	
View Profile	x	x	
Edit Profile	x	x	
View Notifications	x	x	
Create Report	x	x	
Manage System			x
Manage Users			x
Configure API Sources			x
Sync Academic Data			x

3.1.3 Non-Screen Functions

#	Feature	System Function	Description
1	Automated Academic Data Synchronization	Scheduled Job	Scheduled job that automatically retrieves publication metadata directly from external academic APIs (OpenAlex and Crossref). The retrieved data is used for publication search, statistical analysis, and publication trend tracking.
2	Automated Trend Calculation	Scheduled Job	Scheduled job that automatically recalculates publication trend statistics after each data synchronization from external academic APIs. The system analyzes total publications, total citations, Open Access rate, and research topic trends, then updates the data displayed on the dashboard and Trending Topics.

3.2 Entity Relationship Diagram



3.3 Authentication

3.3.1 Register

Role: User

Function trigger: The user clicks Đăng kí.

Function description:

- Purpose: Create a new user account.
- Data processing: Validate user information and store it in the database.

Function Details:

- Whether the username already exists
- Validate password requirements.
- Create a new account.
- Display a success message.
- Sign up and check your email.

Tạo tài khoản ScholarTrend

Tham gia cộng đồng nghiên cứu và khám phá xu hướng khoa học toàn cầu.

Họ và tên

 Nguyễn Văn A

Email

 example@email.com

Tên đăng nhập

 Username

Mật khẩu

 Password

Đăng ký

Đã có tài khoản? [Đăng nhập](#)

- Email already exists

Tạo tài khoản ScholarTrend

Tham gia cộng đồng nghiên cứu và khám phá xu hướng khoa học toàn cầu.

Email đã được đăng ký.

Họ và tên

 Trần Trung Hiếu

Email

 eldenring.674@gamil.com

Tên đăng nhập

 Hizeu

Mật khẩu



- Username already exists.

Tạo tài khoản ScholarTrend

Tham gia cộng đồng nghiên cứu và khám phá xu hướng khoa học toàn cầu.

Tên đăng nhập đã tồn tại.

Họ và tên

 Nguyễn Ngọc Bảo Trâm

Email

 Trammmm@gmail.com

Tên đăng nhập

 baotram

Mật khẩu



Đăng ký

Đã có tài khoản? [Đăng nhập](#)

- Password does not meet the required format.

ScholarTrend

Tạo tài khoản ScholarTrend

Tham gia cộng đồng nghiên cứu và khám phá xu hướng khoa học toàn cầu.

Mật khẩu phải có ít nhất 8 ký tự và bao gồm cả chữ và số.

Họ và tên

 Nguyễn Ngọc Bảo Trâm

Email

 Trammmm@gmail.com

Tên đăng nhập

 baotrammm

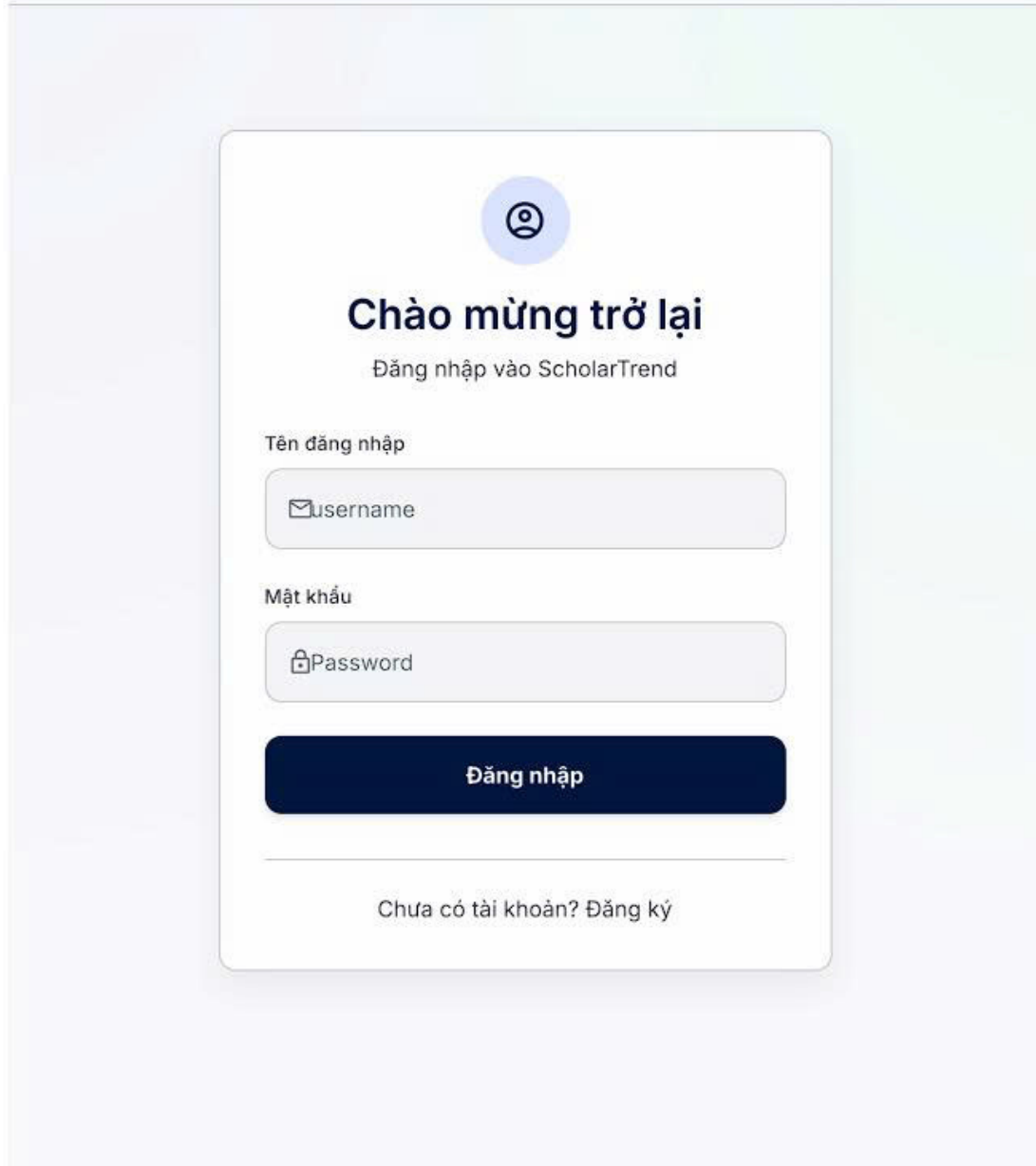
Mật khẩu



Đăng ký

Đã có tài khoản? [Đăng nhập](#)

- Registration successful.

The image shows a login interface for 'ScholarTrend'. At the top, the name 'ScholarTrend' is displayed in a dark blue font. Below this, a light blue circular icon with a white user silhouette is centered. Under the icon, the text 'Chào mừng trở lại' (Welcome back) is written in a large, bold, dark blue font, followed by 'Đăng nhập vào ScholarTrend' (Log in to ScholarTrend) in a smaller, regular dark blue font. The login form consists of two input fields: 'Tên đăng nhập' (Username) with an envelope icon and 'Mật khẩu' (Password) with a lock icon. Both fields are light gray with rounded corners. Below these fields is a dark blue button with the text 'Đăng nhập' (Log in) in white. At the bottom of the form, there is a link that says 'Chưa có tài khoản? Đăng ký' (Don't have an account? Register).

3.3.2 Login

Role: User

Function trigger:

- The user enters a username and password to log in.

Function description:

- **Purpose:** Allow users to access the system.
- **Data processing:** Authenticate the entered username and password against the system database.

Function Details:

- If the login is successful, redirect the user to the Home page.
- If the login fails, display the error message: “Tên đăng nhập hoặc mật khẩu không đúng.”

ScholarTrend



Chào mừng trở lại

Đăng nhập vào ScholarTrend

Tên đăng nhập

username

Mật khẩu

Password

Đăng nhập

Chưa có tài khoản? Đăng ký

- Invalid username or password.

ScholarTrend



Chào mừng trở lại

Đăng nhập vào ScholarTrend

Tên đăng nhập

Mật khẩu

Tên đăng nhập hoặc mật khẩu không đúng.

Đăng nhập

- Login successful.

DỮ LIỆU HỌC THUẬT THỜI GIAN THỰC

Giải mã xu hướng Khoa học toàn cầu

Theo dõi hàng triệu ấn phẩm, phân tích tốc độ phát triển của các lĩnh vực và đón đầu những đột phá nghiên cứu quan trọng nhất.



Tìm kiếm bài báo nghiên cứu, tác giả hoặc từ khóa...

Khám phá ngay →

Chủ đề đang nổi bật

Xem tất cả xu hướng ↗



+21% so với cùng kỳ

Năng lượng Tái tạo

Hiệu suất pin mặt trời thế hệ mới và lưu trữ năng lượng quy mô lớn.

4.758+

Bài báo mới



+17% so với cùng kỳ

Thu giữ Carbon

Công nghệ thu giữ carbon trực tiếp từ không khí và lưu trữ địa chất.

14.186+

Bài báo mới



+7% so với cùng kỳ

Chỉnh sửa Gen CRISPR

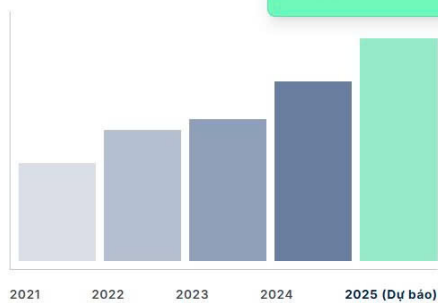
Ứng dụng CRISPR trong điều trị bệnh di truyền và nông nghiệp.

3.977+

Bài báo mới



Tần suất xuất bản theo năm



TỈ LỆ TĂNG TRƯỞNG
21,6%
Trung bình năm

Phân tích xu hướng chuyên sâu

ScholarTrend hỗ trợ thu thập và phân tích metadata bài báo khoa học từ các nguồn học thuật uy tín. Hệ thống giúp người dùng theo dõi xu hướng nghiên cứu, khám phá các chủ đề nổi bật và hỗ trợ quá trình tìm kiếm, phân tích tài liệu khoa học.

- ✓ Theo dõi số lượng bài báo theo từng năm và xu hướng xuất bản.
- ✓ Phân tích các chủ đề nghiên cứu nổi bật dựa trên dữ liệu học thuật.
- ✓ Hiện thị Dashboard trực quan với biểu đồ thống kê và kết quả phân tích.



Đừng bỏ lỡ bất kỳ đột phá nào

Theo dõi các chủ đề nghiên cứu để nhận thông báo khi có bài báo mới hoặc khi xu hướng xuất bản của chủ đề thay đổi.

3.3.3 Profile Management

Update Profile

Role: User

Function trigger: The user opens the Profile page and clicks **Chỉnh sửa thông tin**.

Function description:

- Purpose: Allow users to update their personal information.
- Data processing: Validate and save the updated information.

Function Details:

- Update name, Email and other personal information.
- Display a success message after saving.

ScholarTrend

Thông tin tài khoản
Quản lý hồ sơ cá nhân và cài đặt bảo mật của bạn.

HỌ VÀ TÊN
Trần Trung Hiếu

EMAIL
eldenring.674@gamil.com

TÊN ĐĂNG NHẬP
Hizeu

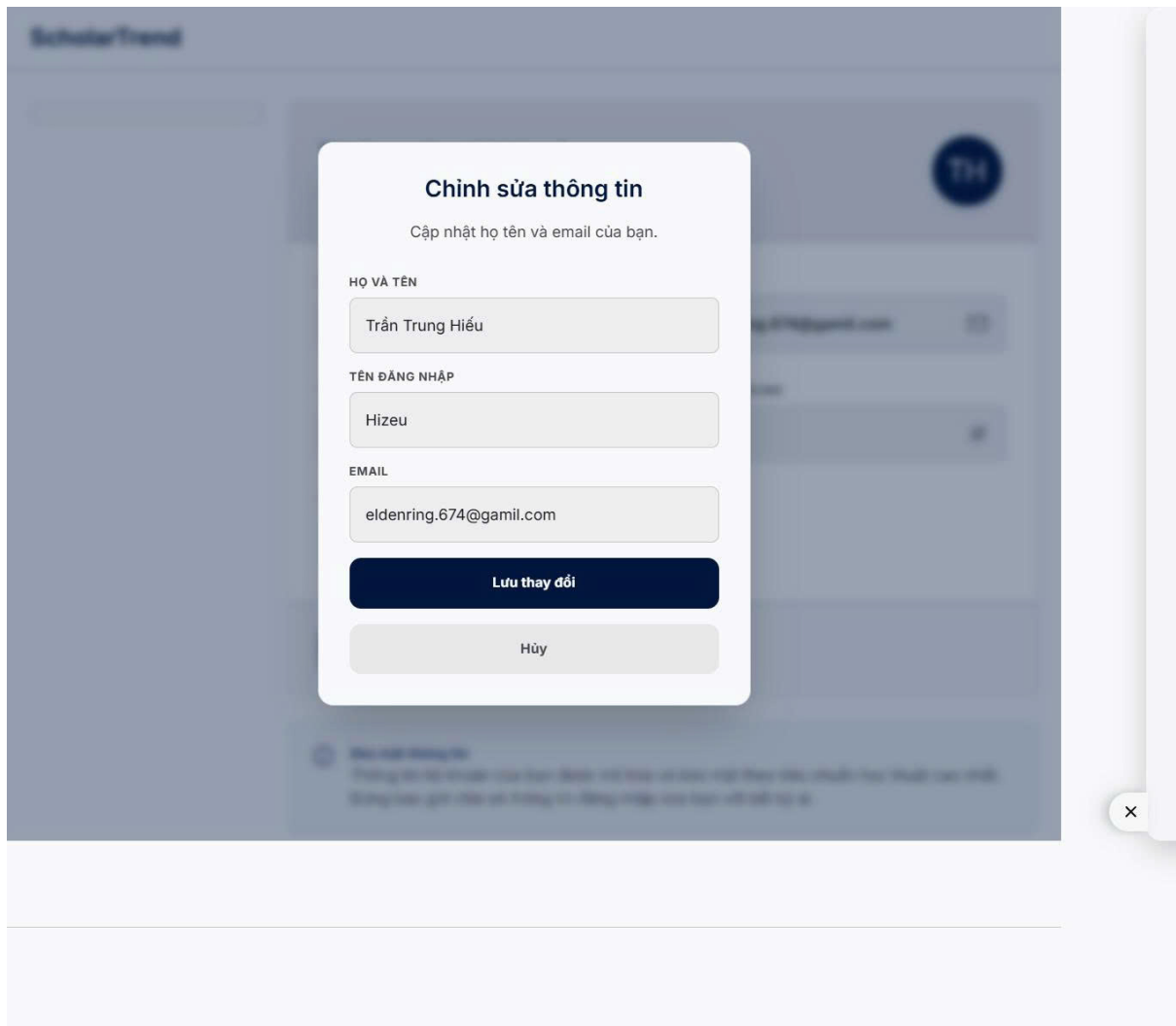
MÃ NGƯỜI DÙNG
31

TRANG THÁI EMAIL
Đã xác minh

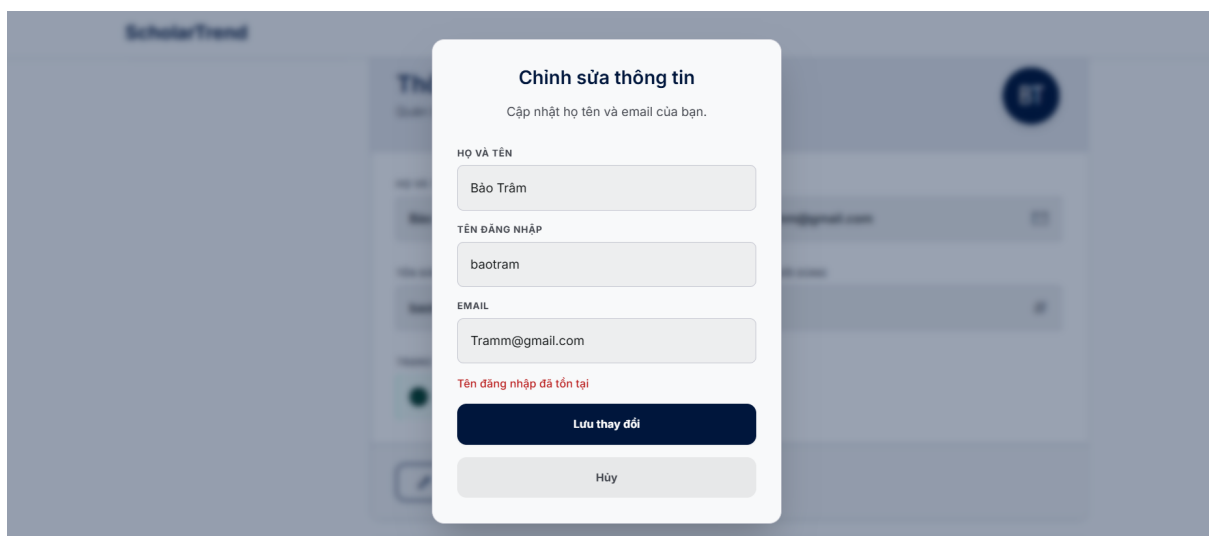
Chỉnh sửa thông tin **Đăng xuất**

Bảo mật thông tin
Thông tin tài khoản của bạn được mã hóa và bảo mật theo tiêu chuẩn học thuật cao nhất.
Đừng bao giờ chia sẻ thông tin đăng nhập của bạn với bất kỳ ai.

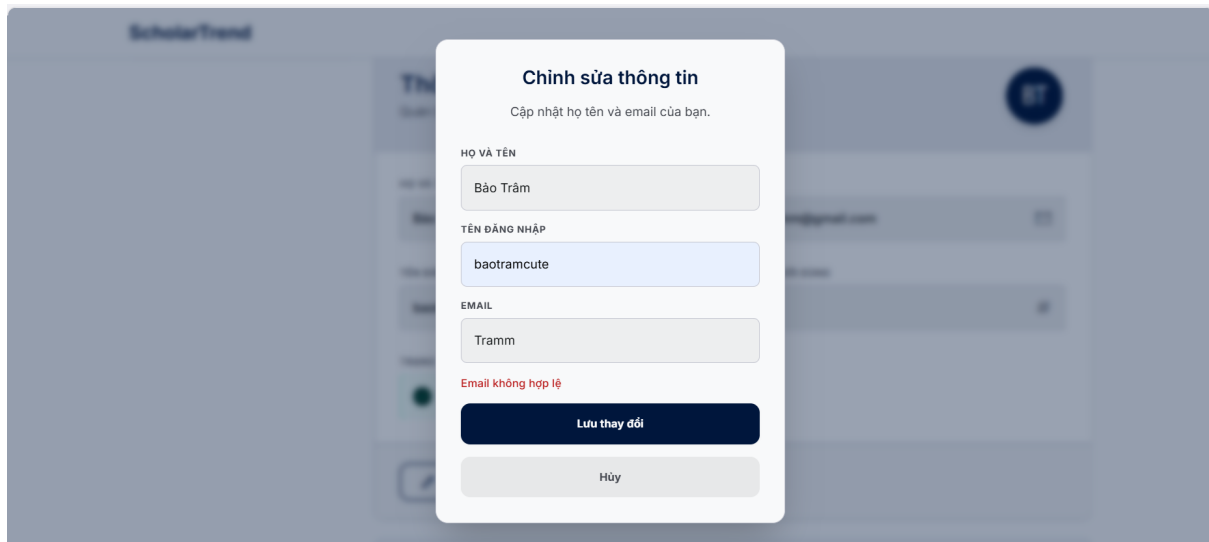
- Edit Profile



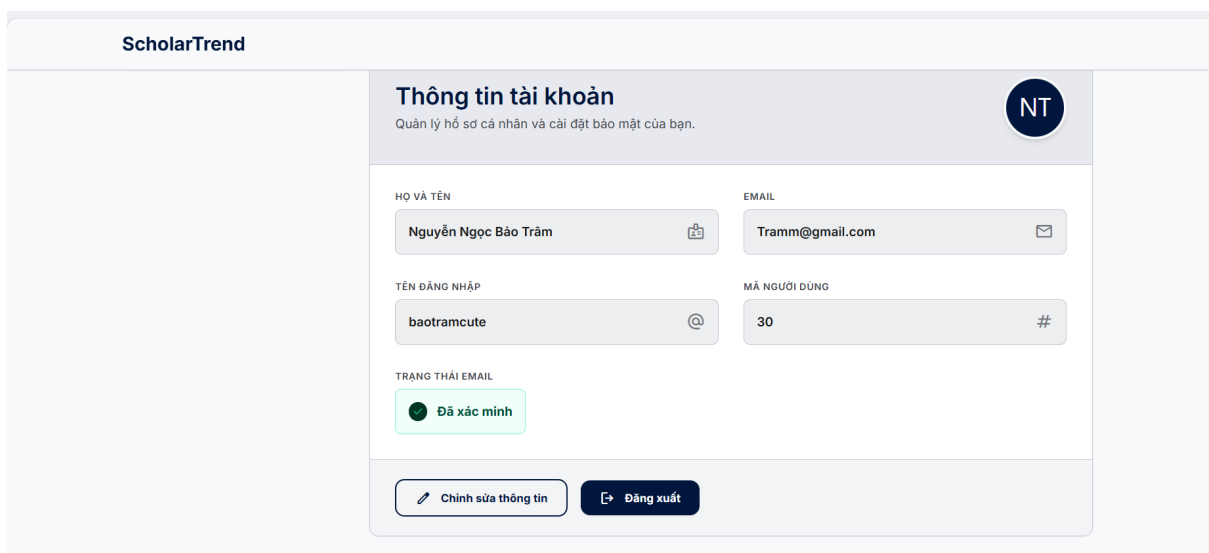
- Username already exists.



- Invalid email address



- Update successful.



3.3.4 Logout

Role: User

Function trigger: The user opens the Profile page and clicks Đăng xuất.

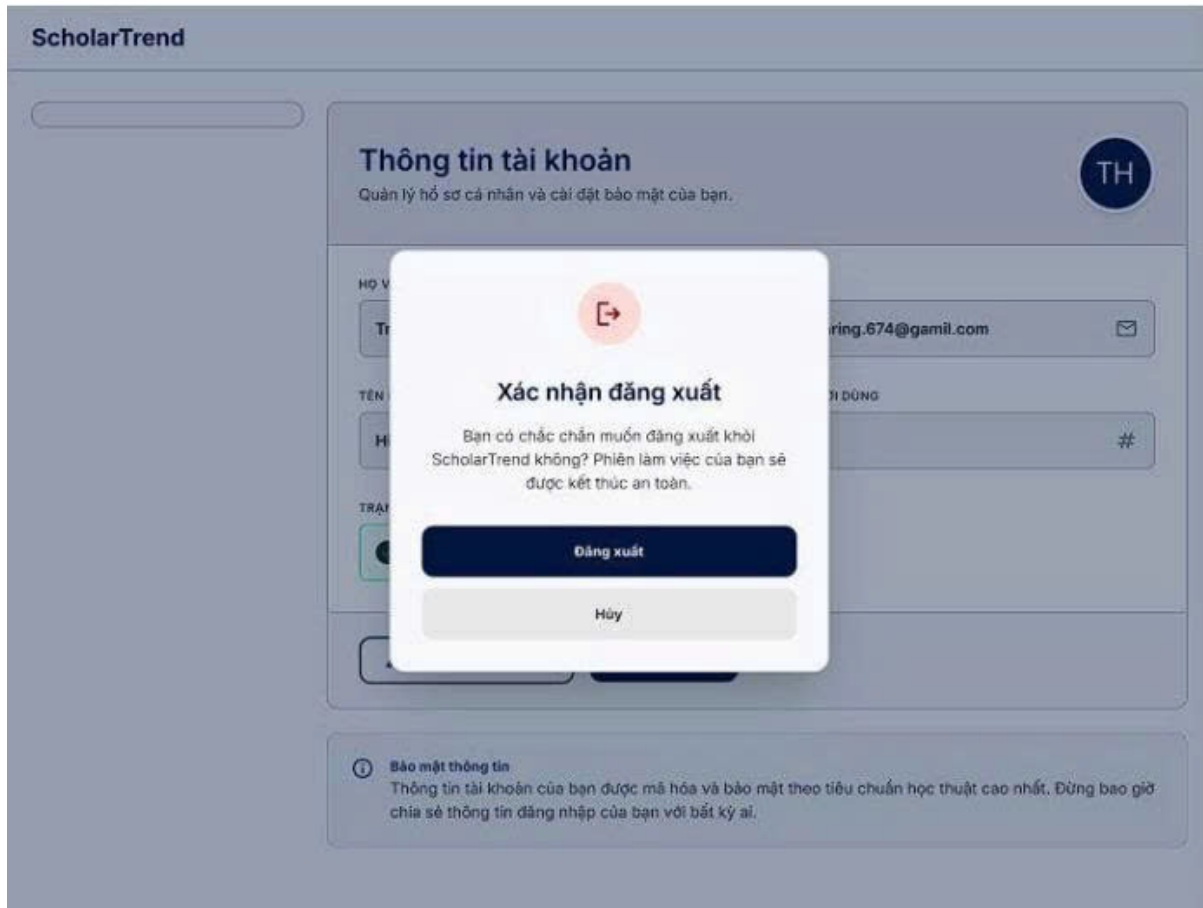
Function description:

- Purpose: Allow the user to securely log out of the system.
- Data processing: End the current user session.

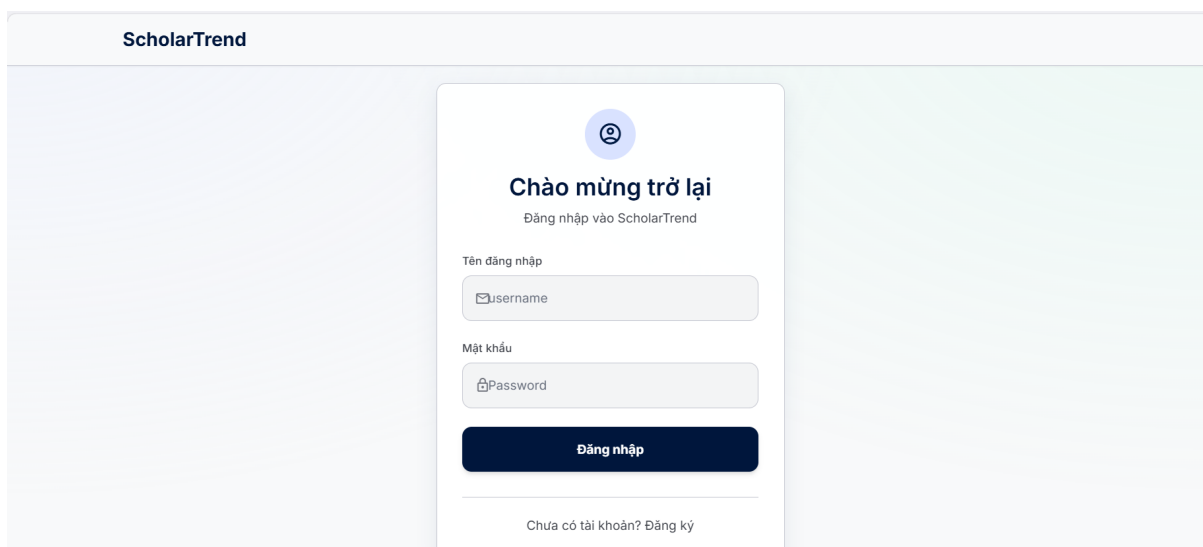
Function Details:

- Display a confirmation dialog asking whether the user wants to Đăng xuất.

- If the user selects Đăng xuất, the system logs the user out and redirects them to the Đăng nhập page.
- If the user selects Hủy, the dialog is closed and the user remains on the current page.



- Signed out successfully.



3.3.5 Search Papers

Role: User

Function Trigger

User opens “*Tìm kiếm*” → Enters a keyword → Selects filters (Publication Year, Open Access, Paper Type) → Clicks **Search**.

Function Description

- **Purpose:**

Allow users to search scientific papers from academic databases based on keywords and filtering conditions.

- **Data Processing:**

Validate the search input, send the request to academic databases (e.g., Crossref API), retrieve matching publications, apply selected filters, and display the search results.

Function Details

- **Validation:**

- Keyword cannot be empty.
- Start year must not be greater than End year.

- **Business Rules:**

- Search supports Title, Keyword and DOI.
- Users may filter by publication year.
- Users may filter by paper type.
- Users may choose to display only Open Access publications.
- Results are displayed with pagination (10 items per page).
- Each result includes Title, Authors, Year, DOI and Citation Count (if available), with the option to save the paper to Favorites.

- **Normal Case:**

- User enters a keyword and selects filters.
- User clicks the **Search** button.
- System queries the academic database.
- Matching publications are returned and displayed.
- User can view paper details, open the DOI link, or save the paper to Favorites.

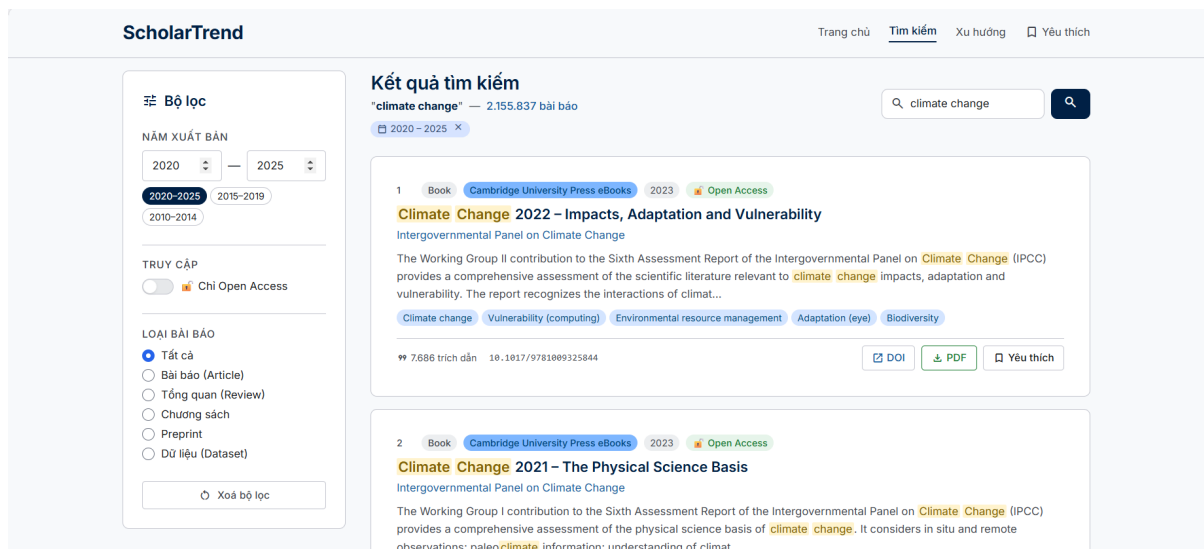
- **Abnormal Case:**

- No papers found → Display “*No matching papers found.*”

- Invalid year range → Request the user to correct the input.
- API unavailable → Display “Unable to retrieve publication data. Please try again later.”

- The start year cannot be greater than the end year.

- Search Results



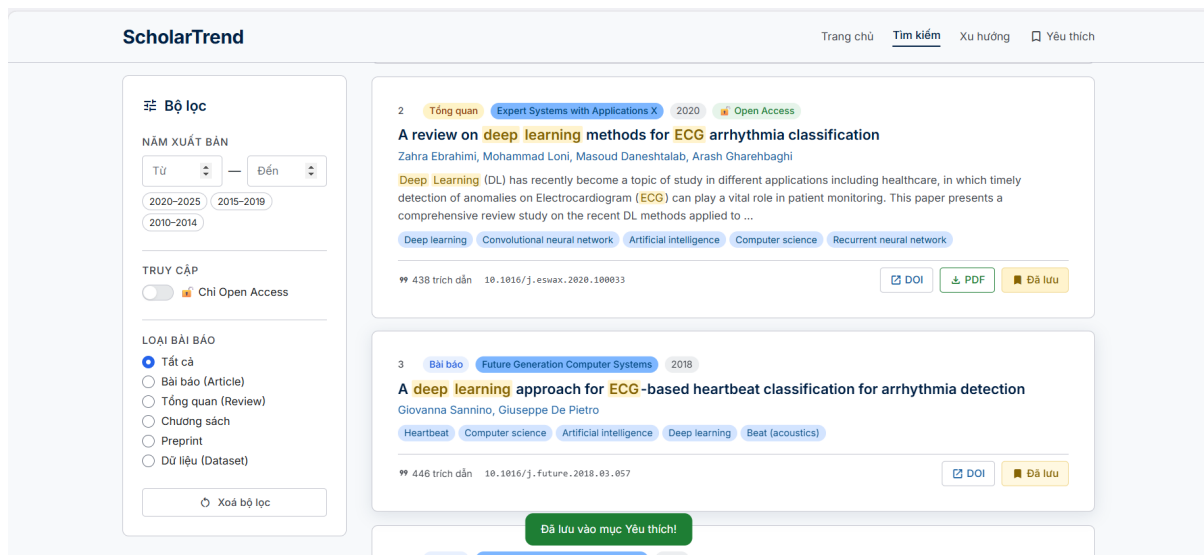
3.3.6 Bookmark

The Favorite Papers feature allows users to save publications of interest for future reference and easy access.

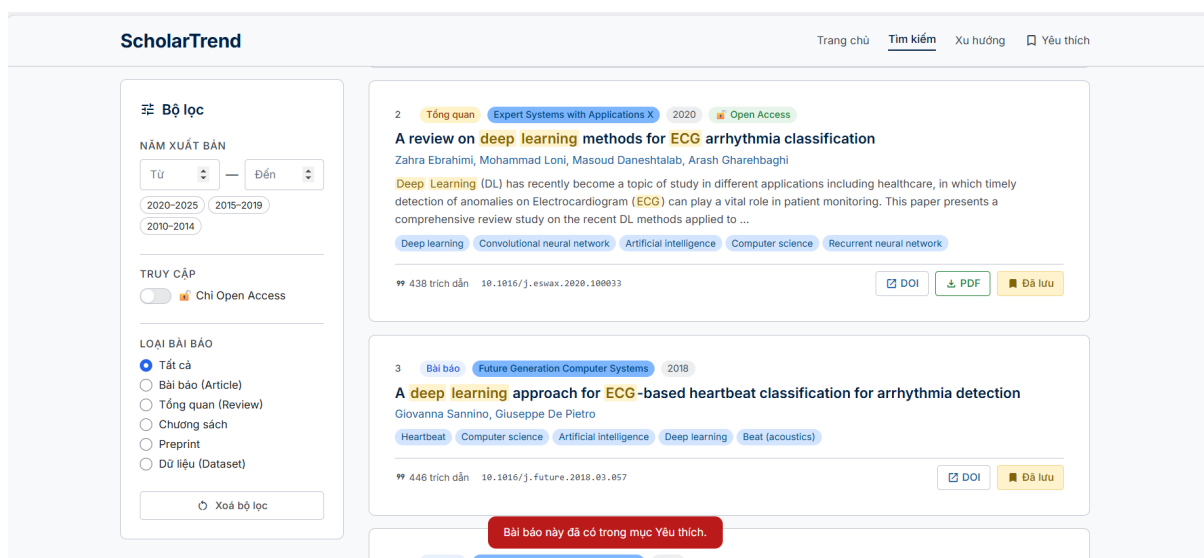
- Users can click the Favorite button on a paper in the Search Results page to add it to their favorite list.
- If the paper is successfully added, the system displays the message "Added to Favorites successfully." and updates the button status to Saved.
- If the selected paper already exists in the favorite list, the system displays the message "This paper is already in your Favorites." and prevents duplicate entries.
- Users can access the Favorites page from the navigation bar to view all saved papers.
- Each saved paper displays the following information:
 - Publication title.
 - Publication year.
 - Data source (e.g., OpenAlex).
 - Saved date and time.
 - DOI button for accessing the original publication.
 - Delete button for removing the paper from the favorite list.
- Users can remove a paper from the favorite list by clicking the Delete button. The system immediately updates the favorite list.
- Users can click the Find More Papers button to return to the search page and continue searching for publications.

Expected Result

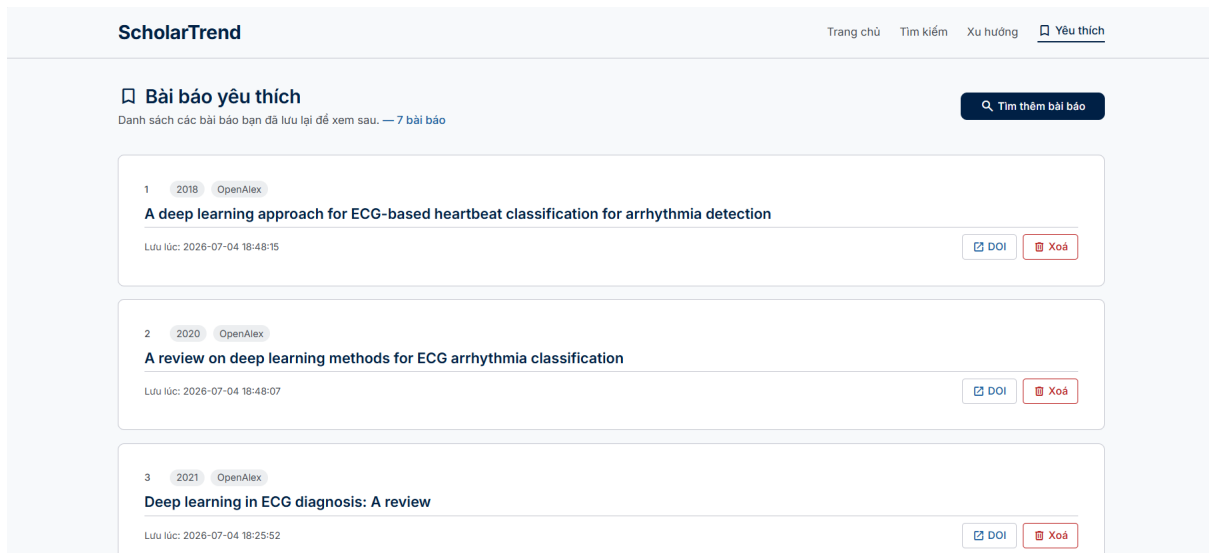
- Each paper can only be saved once in the favorite list.
- Users can easily manage, review, and access their saved publications.
- All changes to the favorite list are synchronized with the database and reflected immediately in the user interface.



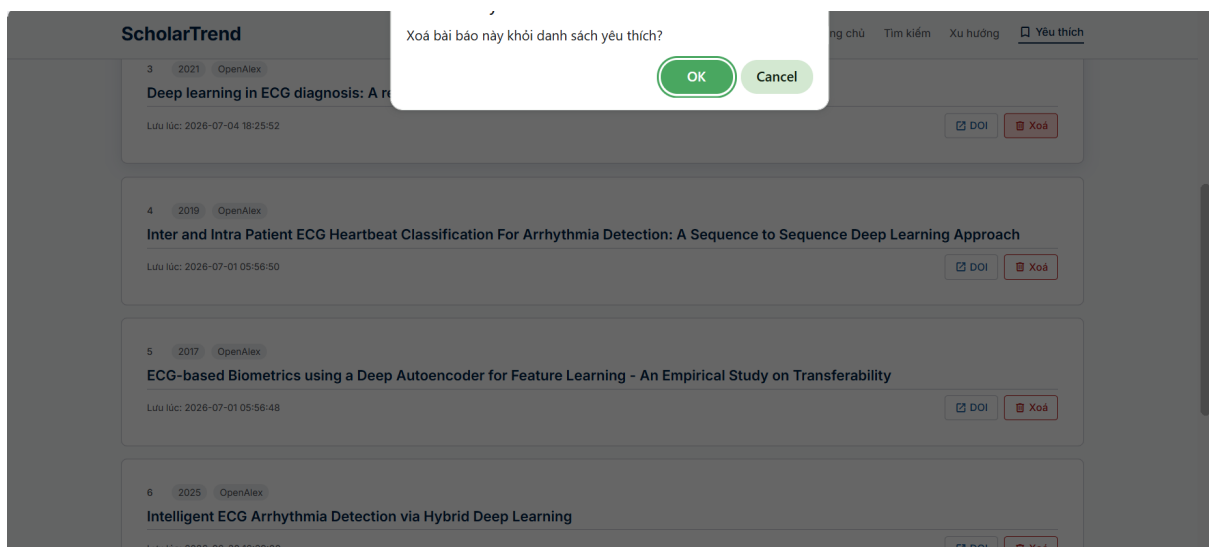
- Marked as Favorite.



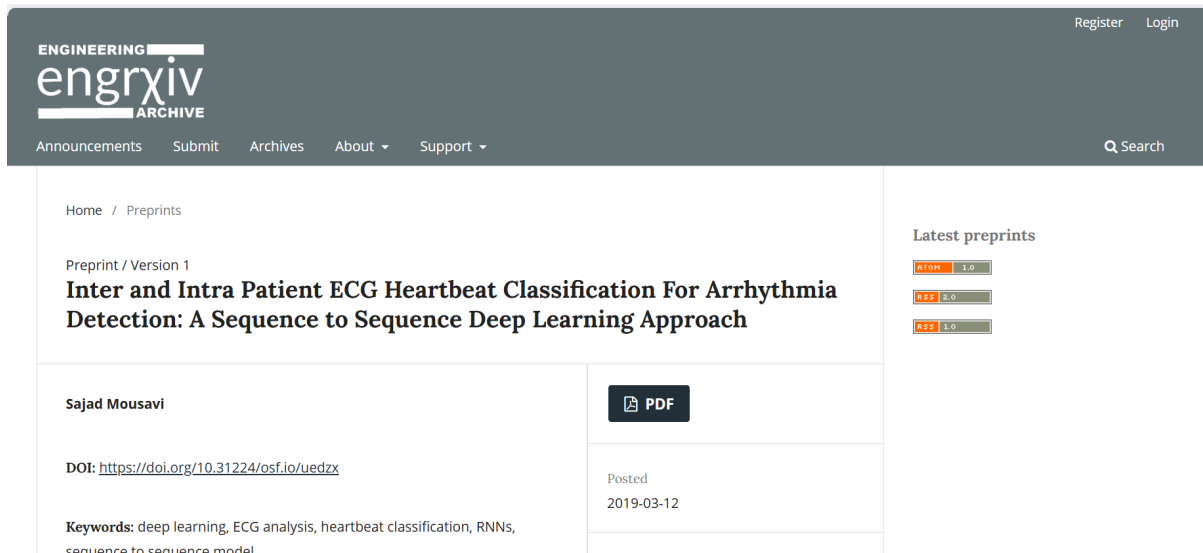
- Bookmarked List



- Remove from Bookmarks.



- Link to the bookmarked paper.



3.3.7 Receive Notifications

3.3.8 Research Trend Analysis

Function Trigger

User opens *Research Trend Analysis* from the main menu.

Function Description

Actors: Researcher, Lecturer, Student

Purpose

Allow users to analyze research trends of a specific research topic within a selected time range.

Data Processing

- Validate the input topic and time range.
- Retrieve publication data from the database.
- Aggregate statistical information.
- Calculate research trend metrics.
- Generate analytical charts.
- Display the research trend dashboard.

Interface

The **Research Trend Dashboard** includes:

- Search bar for entering a research topic.

- Time range selector.
- **Analyze** button.
- Two tabs:
 - Dashboard
 - Trending Topics
- Dashboard widgets:
 - Total Publications
 - Total Citations
 - Open Access Ratio
 - Trend Growth
- Charts:
 - Publications by Year
 - Top Keywords
 - Top Journals
 - Top Authors
 - Featured Publications

Function Details

Validation

- Topic keyword cannot be empty.
- Start Year must not be greater than End Year.
- The selected time range must be within the supported system range.

Business Rules

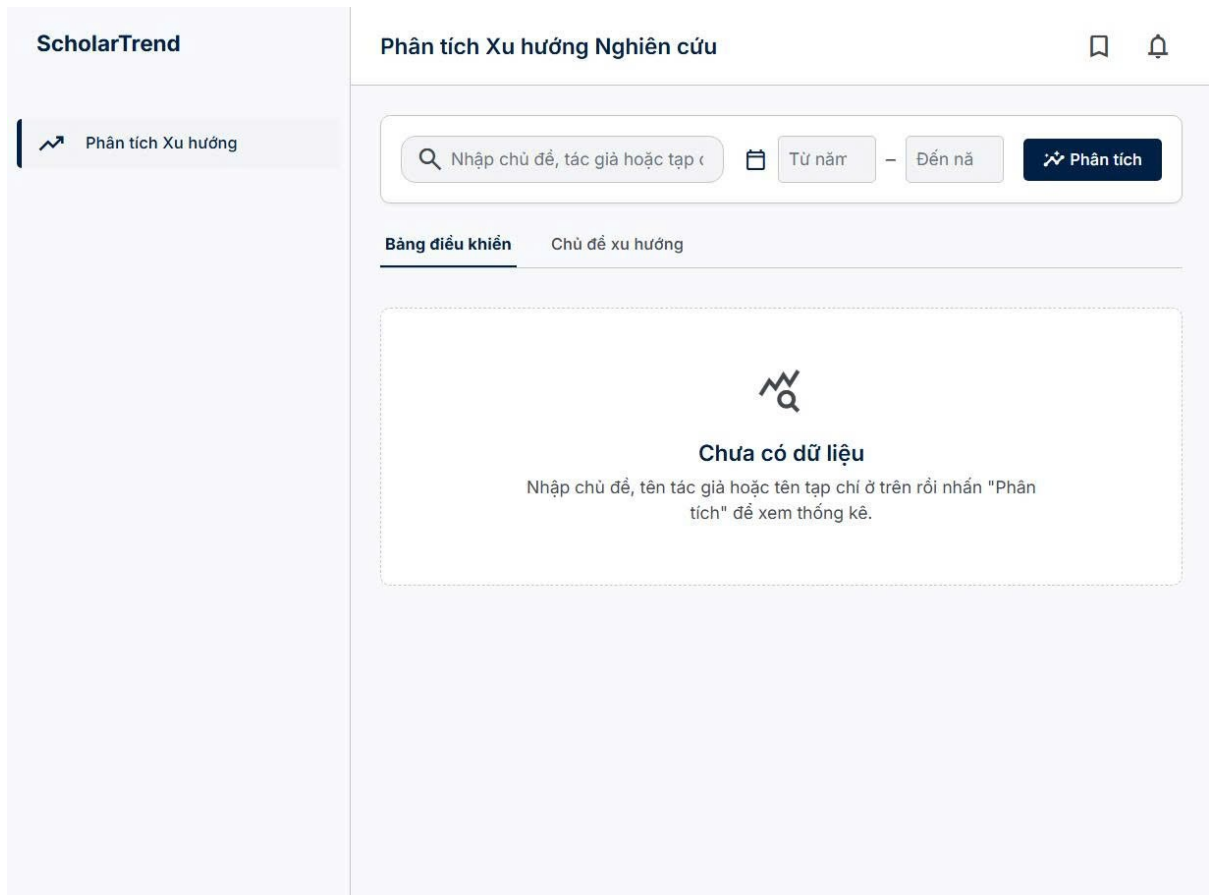
- Statistics are calculated based on publications retrieved by the system.
- Trend Growth is calculated by comparing the selected period with the previous period of the same duration.
- Users can follow or unfollow a trending topic.
- Dashboard data is refreshed whenever the topic or time range changes.
- Trending Topics are ranked according to publication growth and research activity.
- Charts and statistics always correspond to the currently selected topic and time range.

Normal Flow

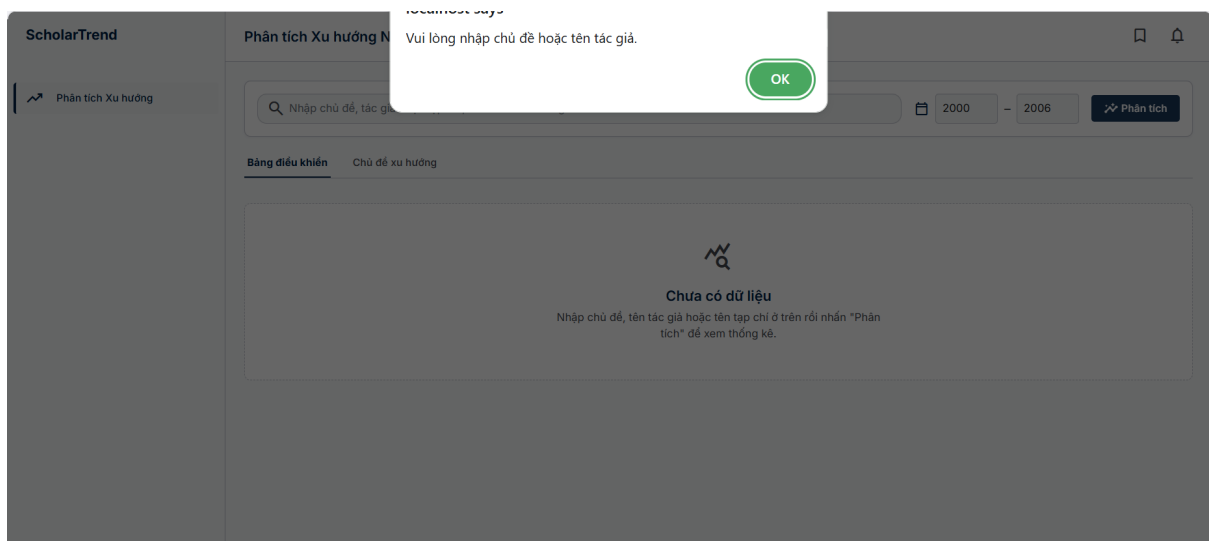
1. User enters a research topic.
2. User selects a time range.
3. User clicks the **Analyze** button.
4. The system validates the input.
5. The system retrieves publication data.
6. The system calculates statistics and generates charts.
7. The dashboard displays the analysis results.
8. User switches to the **Trending Topics** tab.
9. User follows or unfollows a trending topic.

Alternative / Exception Flow

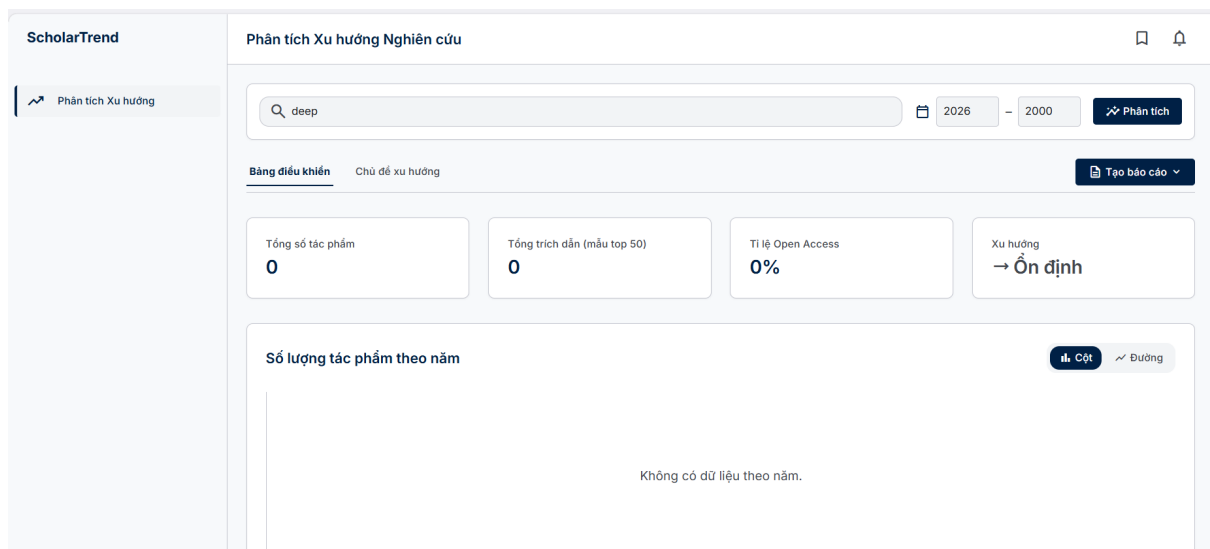
- No matching data found.
 - Display message: *"No data available."*
- Invalid time range.
 - Request the user to correct the selected time range.
- System or database error.
 - Display message: *"Unable to load data. Please try again later."*



- Please enter a search keyword.



- The start year cannot be greater than the end year.



- Analysis Results.

 khoa học

2020

– 2026

 Phân tíchBảng điều khiển

Chủ đề xu hướng

 Tạo báo cáo ▾

Tổng số tác phẩm

59K

Tổng trích dẫn
(mẫu top 50)

13K

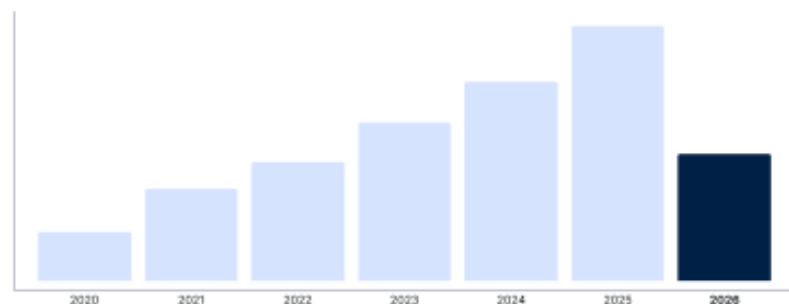
Tỉ lệ Open
Access

98%

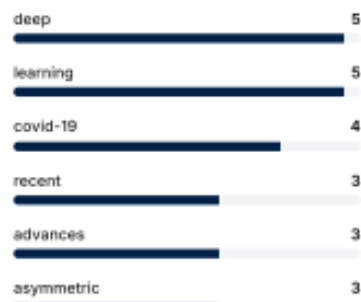
Xu hướng

 Tăng
(137%)

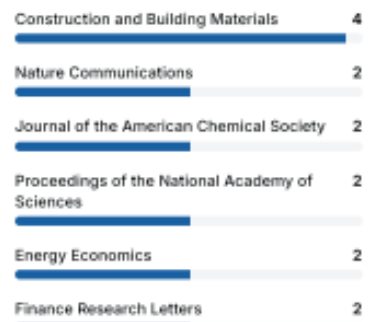
Số lượng tác phẩm theo năm

 Cột Đường

Phân bố chủ đề con



Phân bố tạp chí



Tác giả nổi bật

TÁC GIẢ	SỐ BÀI	TRÍCH DẪN
Sudabeh Alatab	1	2K
Sadaf G Sepanlou	1	2K
Kevin Ikuta	1	2K
Homayoon Vahedi	1	2K

ScholarTrend

Phân tích Xu hướng

TÁC GIẢ	SỐ BÀI	TRÍCH DẪN
Catherine Bisignano	1	2K
Saeid Safiri	1	2K
Anahita Sadeghi	1	2K
Molly R Nixon	1	2K

Tác phẩm nổi bật

The global, regional, and national burden of inflammatory bowel disease in 195 countries and territories, 1990–2017: a systematic analysis for the Global Burden of Disease Study 2017 **OPEN ACCESS**

2020 · 2K trích dẫn

[Xem DOI →](#)

Deep learning in cancer diagnosis, prognosis and treatment selection **OPEN ACCESS**

2021 · 809 trích dẫn

[Xem DOI →](#)

Spiking Neural Networks and Their Applications: A Review **OPEN ACCESS**

2022 · 570 trích dẫn

[Xem DOI →](#)

3.3.9 Export Research Trend Report

Function Trigger User opens *Research Trend Analysis* → Enters a research topic and selects a time range → Clicks Analyze → Clicks Generate Report → Selects an export option (Preview & Print / Save PDF, Download PDF, or Download HTML). Function Description Actors:

Researcher, Lecturer, Student Purpose Allow users to export research trend analysis results into different report formats for printing, sharing, or offline viewing. Data Processing

- Verify that research trend analysis has been completed.
- Validate that report data is available.
- Collect the current analysis results, including:
 - Statistical summaries
 - Charts and visualizations
 - Trending Topics
 - Top Journals

- Top Authors
- Top Keywords
- Featured Publications
- Generate the report in the selected format.
- Prepare the report for preview or download.

Function Details Validation

- Research trend analysis must be completed before exporting.
- Report data must be available.

Business Rules

- The exported report contains the currently displayed analysis results.
- Users may choose one of the following export options:
 - Preview & Print / Save PDF
 - Download PDF
 - Download HTML
- Charts, statistical summaries, and publication information are included in the exported report.
- The exported report reflects the currently selected research topic and time range.
- The report preserves the dashboard layout and chart formatting.

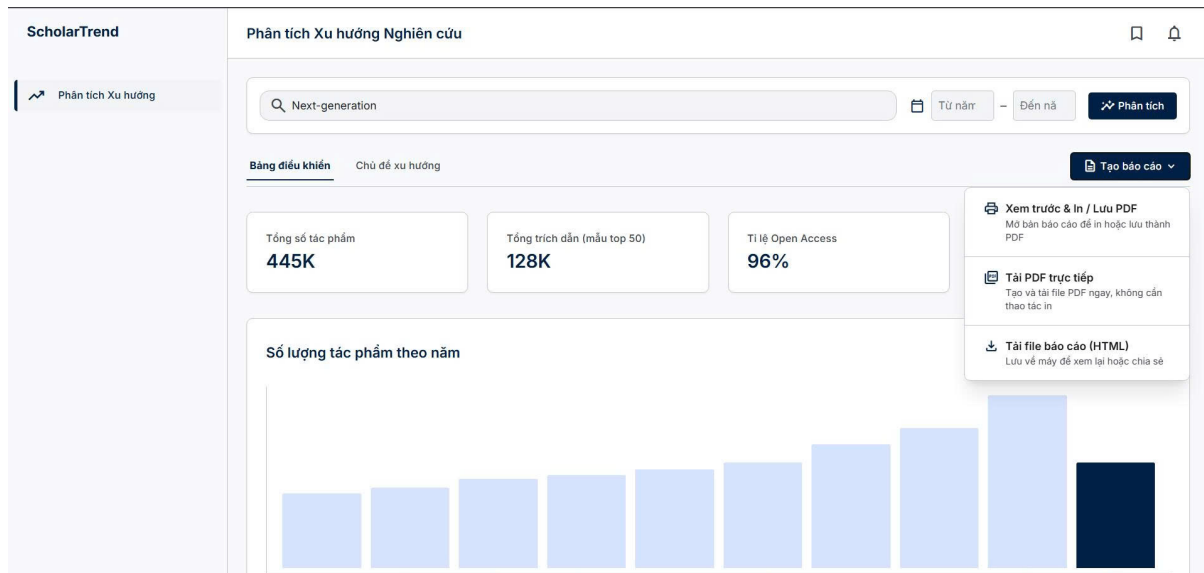
Normal Flow

1. User completes the research trend analysis.
2. User clicks the Generate Report button.
3. User selects an export format.
4. The system validates the report data.
5. The system generates the report.
6. The report is previewed or downloaded successfully.

Alternative / Exception Flow

- No analysis data available.
 - Display message: *"No report data available."*

- Report generation fails.
 - Display message: *"Unable to generate report. Please try again later."*
- Selected export format is unavailable.
 - Display message: *"Selected export format is unavailable."*



- Export Report.

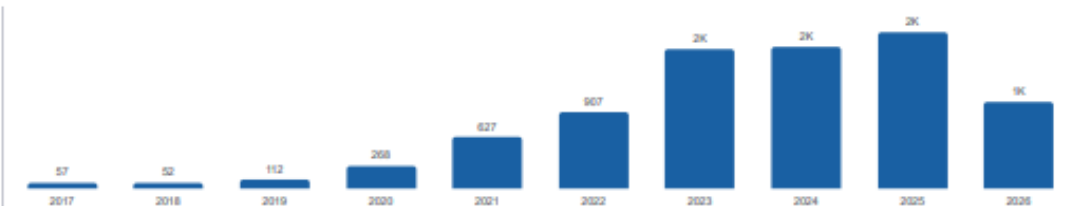
BÁO CÁO PHÂN TÍCH XU HƯỚNG NGHIÊN CỨU

Chủ đề / tác giả tìm kiếm	y tế
Khoảng thời gian	Toàn bộ thời gian
Ngày tạo báo cáo	4/7/2026 19:44

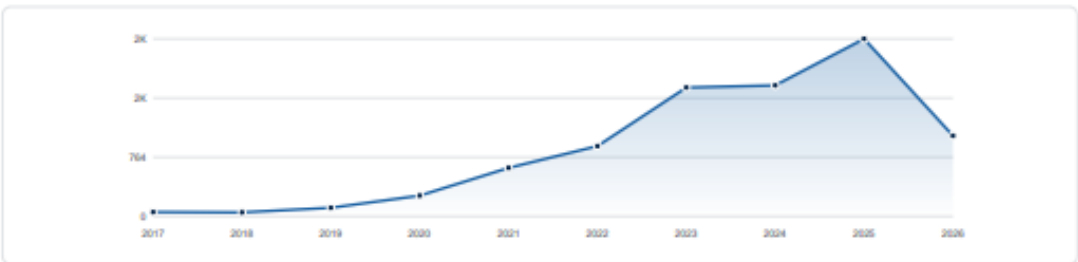
Tổng quan

Tổng số tác phẩm 9K	Tổng trích dẫn (mẫu top 50) 4K	Tỉ lệ Open Access 98%	Xu hướng Tăng trưởng (580%)
-------------------------------	--	---------------------------------	---------------------------------------

Số lượng tác phẩm theo năm (biểu đồ cột)



Biểu đồ đường xu hướng theo thời gian



Phân bố chủ đề con

connectedness	9
between	9
impact	8
spillovers	8
green	6
asymmetric	4

Phân bố tạp chí

Finance Research Letters	4
Economic Analysis and Policy	4
Research in International Business and Finance	3
Energy Economics	2
Sustainable Production and Consumption	2
The International Journal of Management Education	2

3.3.10 Follow Topic

Function Activation

Users open the **Trend Analysis** page → Enter a research topic and select the time range
→ Click **Analyze** → Click the **Follow** button to follow a research topic.

Description

Actors: Researchers, Lecturers, Students.

Purpose

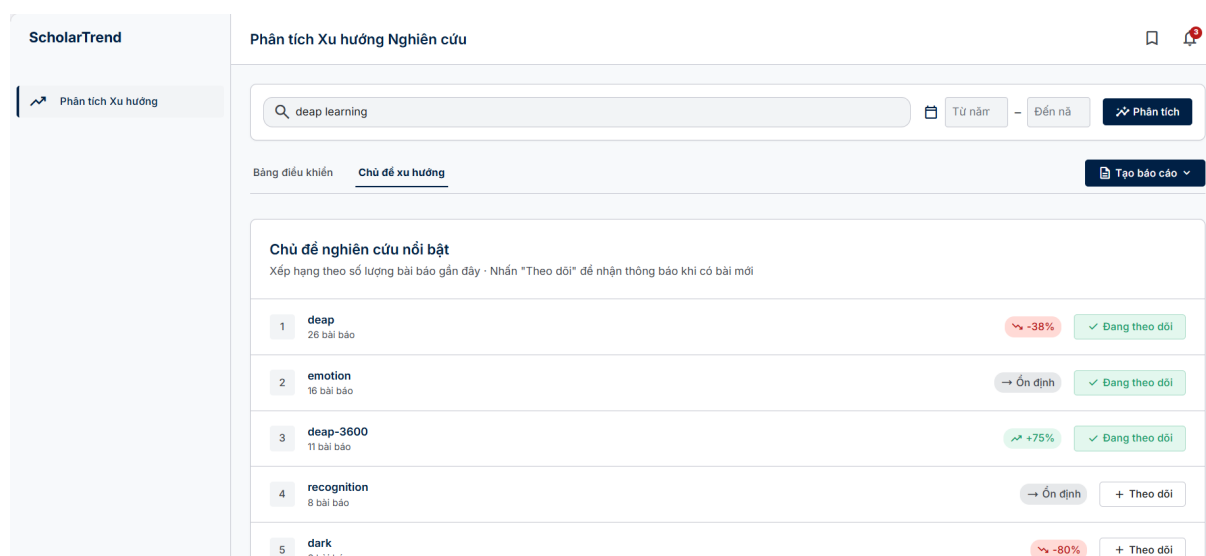
Allow users to follow research topics of interest for easier management and quick access.

Data Processing

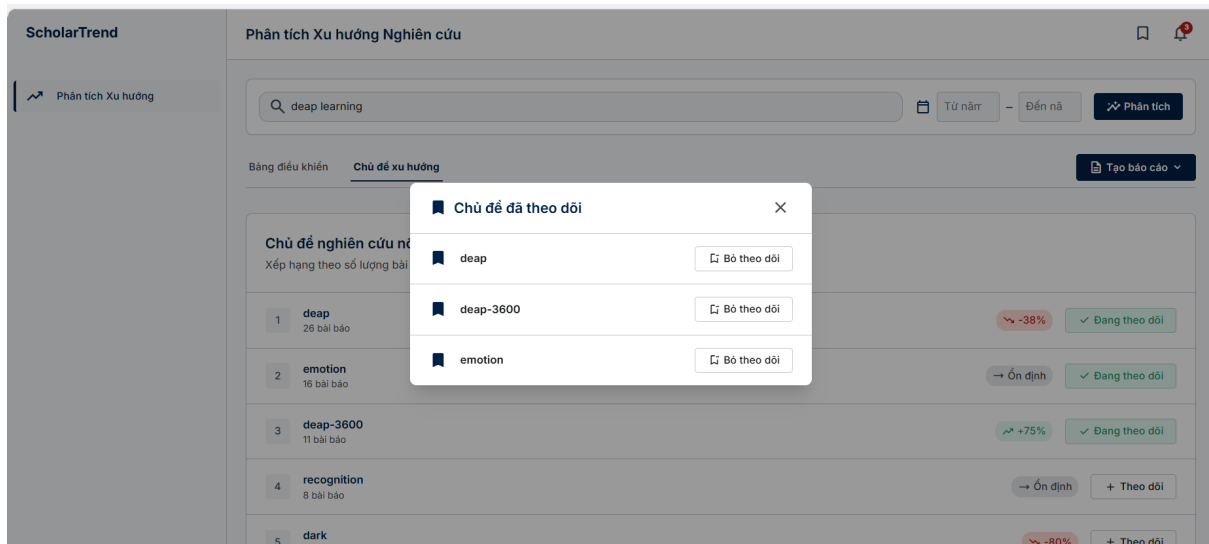
- Verify that the user is logged into the system.
- Check whether the selected topic has already been followed.
- If the topic has not been followed, save it to the user's followed topic list and update its status to **Following**.
- If the topic is already followed, prevent duplicate entries.
- Allow users to view the list of followed topics.
- Allow users to unfollow a topic by clicking the **Unfollow** button.
- Update the followed topic list after each follow or unfollow operation.

Result

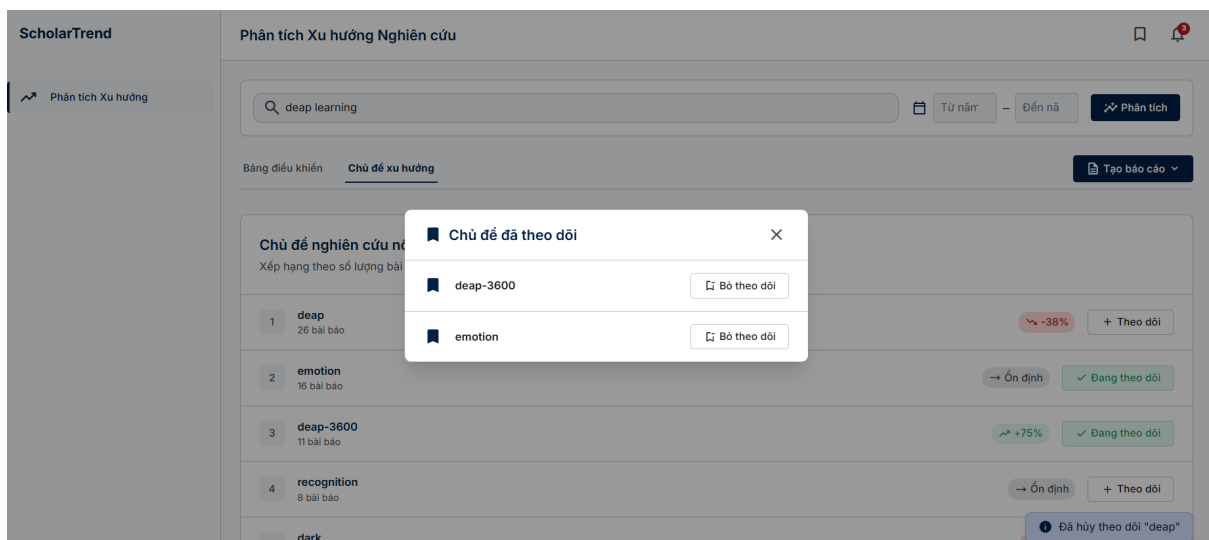
- The selected topic is added to the followed topic list.
- The followed topic list is updated correctly.
- Users can follow or unfollow topics at any time.



- Follow topics list



- Unfollow topic



3.3.11 Receive Notifications

Function Activation

Users log in to the system. Notifications are automatically displayed when new notifications are available.

Description

Actors: Researchers, Lecturers, Students.

Purpose

Allow users to receive notifications about followed research topics and system announcements from administrators.

Data Processing

- Verify that the user is logged into the system.
- Retrieve unread notifications from the database.
- Display notifications related to followed research topics.
- Display announcements and notifications sent by administrators.
- Allow users to mark notifications as read.
- Update the notification status after it has been read.

Result

- Users receive notifications about followed topics.
- Users receive announcements from administrators.
- Notification status is updated correctly after being read.

The screenshot displays the ScholarTrend application. The main content area is titled "Phân tích Xu hướng Nghiên cứu" (Research Trend Analysis). It features a search bar with the text "kinh tế" (economy). Below the search bar, there's a section for "Chủ đề xu hướng" (Trending Topics) with a table listing five topics:

Rank	Topic	Count	Change	Action
1	kinh	17 bài báo		
2	động	11 bài báo		
3	tăng	10 bài báo	+133%	+ Theo dõi
4	phát	6 bài báo	+400%	+ Theo dõi
5	trường	5 bài báo	+300%	+ Theo dõi

On the right side, a "Thông báo" (Notification) overlay is visible, showing three notifications about following topics: "kinh", "động", and "deep-3600". Each notification includes a bell icon, a title, a description, and a "Vừa xong" (Just now) timestamp. At the bottom of the overlay, there's a link to "Xem tất cả chủ đề đã theo dõi" (View all followed topics).

- Mark as Read.

ScholarTrend

Phân tích Xu hướng

Phân tích Xu hướng Nghiên cứu

🔍 kinh tế

Bảng điều khiển

Chủ đề xu hướng

Chủ đề nghiên cứu nổi bật

Xếp hạng theo số lượng bài báo gần đây · Nhấn "Theo dõi" để nhận thông báo khi có bài mới

1	kinh	17 bài báo		
2	động	11 bài báo		
3	tăng	10 bài báo	+133%	+ Theo dõi
4	phát	6 bài báo	+400%	+ Theo dõi
5	trường		+300%	+ Theo dõi

Thông báo

Đánh dấu đã đọc

🔔 Đã theo dõi chủ đề mới

Bạn đang theo dõi "động". Bạn sẽ nhận thông báo khi có bài mới hoặc xu hướng thay đổi.

Vừa xong

🔔 Đã theo dõi chủ đề mới

Bạn đang theo dõi "kinh". Bạn sẽ nhận thông báo khi có bài mới hoặc xu hướng thay đổi.

Vừa xong

🔔 Đã theo dõi chủ đề mới

Bạn đang theo dõi "deap-3600". Bạn sẽ nhận thông báo khi có bài mới hoặc xu hướng thay đổi.

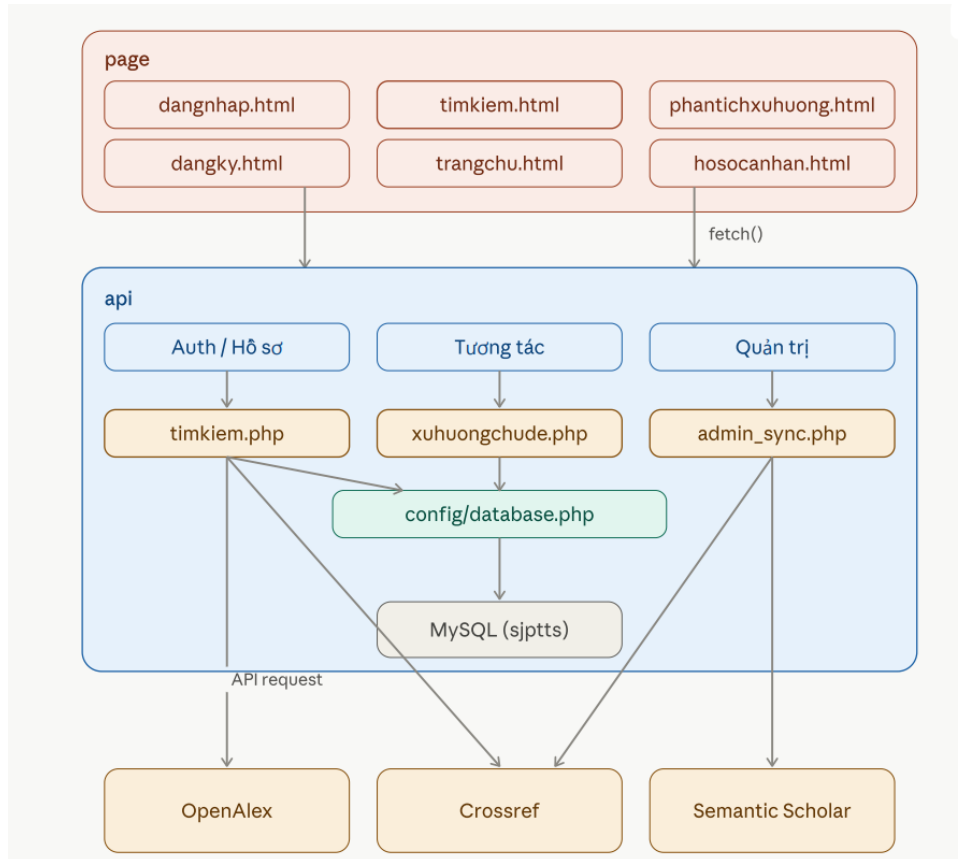
Vừa xong

Xem tất cả chủ đề đã theo dõi

IV. Software Design Description

1 System Design

1.1 System Architecture



1.1.1 Frontend App

The frontend application is developed using HTML, CSS, and JavaScript to provide the user interface for interacting with the system. It is responsible for presenting data, handling user interactions, and communicating with the backend system through HTTP requests.

- **Pages:** Responsible for rendering the main user interfaces, including Login Page, Dashboard Page, Publication Search Page, Trend Analysis Page, Bookmark Page, and Notification Page.
- **Components:** Reusable user interface elements used to build application pages, such as search bars, filter panels, publication cards, charts, tables, and navigation menus.
- **Service:** Responsible for managing communication between the frontend and backend system through HTTP requests and responses, including sending requests and processing returned data.

1.1.2 Backend System

The backend system follows a layered architecture consisting of Controller Layer, Service Layer, and Repository Layer. This architecture improves maintainability, scalability, and separation of concerns.

- **Controller Layer:** Handles incoming HTTP requests from the frontend, processes request validation, and forwards requests to the appropriate service layer.
- **Service Layer:** Contains business logic for processing publication data, trend analysis, user personalization, and system administration.
- **Repository Layer:** Responsible for interacting with the database, executing queries, and managing data persistence.

1.1.3 External Systems

The system integrates with external services to retrieve academic publication data and store system information.

- **Academic APIs:** External academic data sources such as Semantic Scholar API, OpenAlex API, and Crossref API are used to retrieve publication metadata.
- **MySQL Database:** Stores user information, publication metadata, bookmarks, followed topics, notifications, and system configuration data.

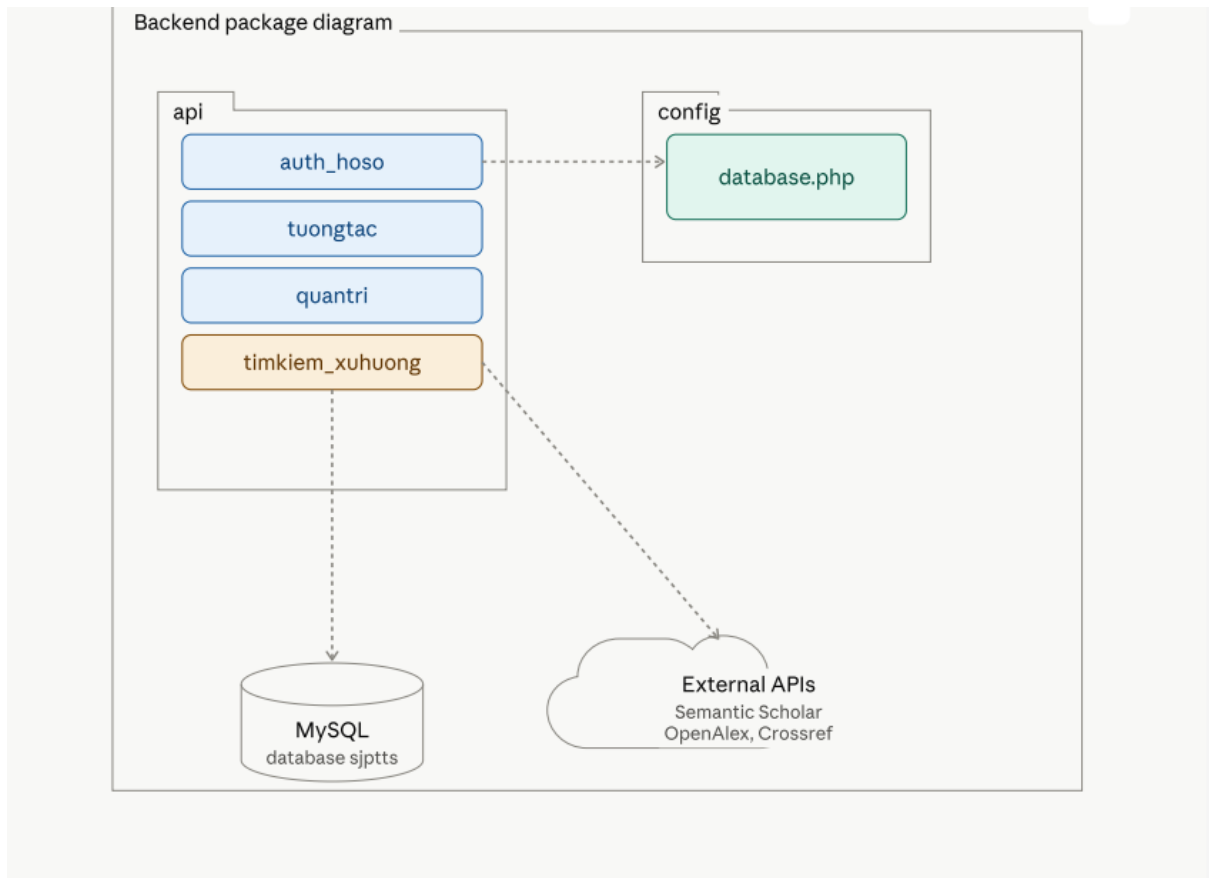
1.2 Package Diagram

1.2.1 Front End



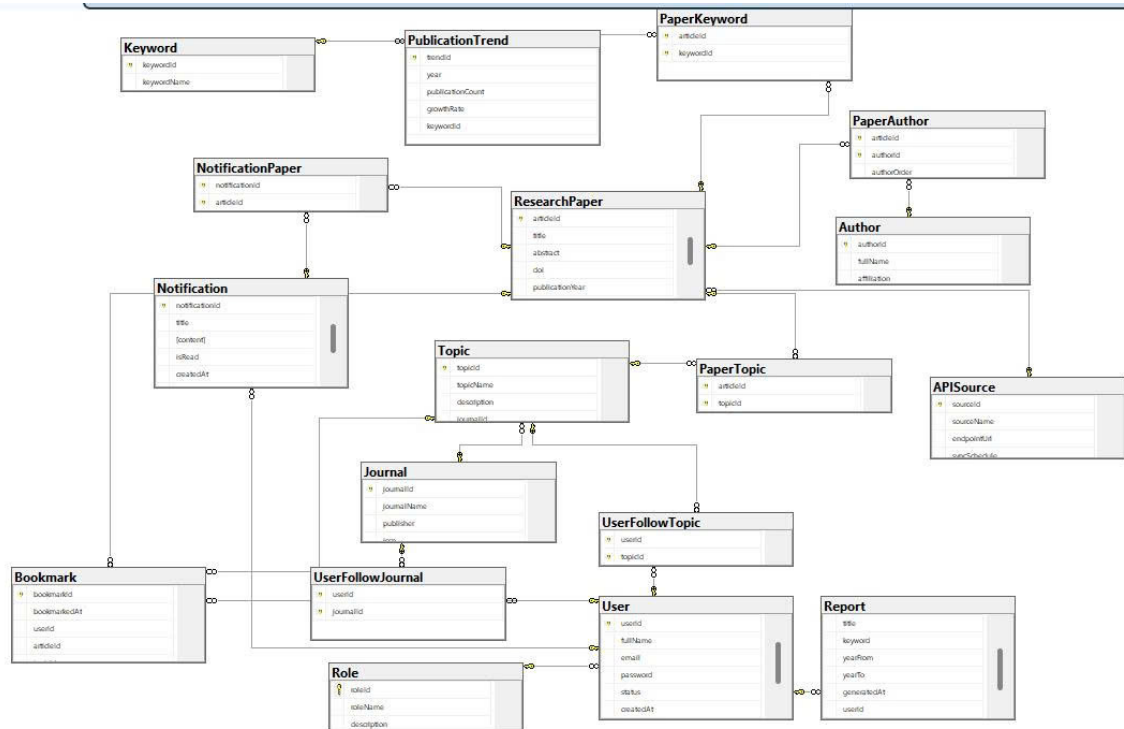
No	Page	Description
1	dangnhap.html, dangky.html	Login and registration pages, calling dangnhap.php / dangky.php.
2	trangchu.html	Public landing page (before login) — hero banner introducing ScholarTrend, a search box for papers/authors/keywords, and a list of trending topics.
3	trangchu1.html	Home page for logged-in users — full navigation bar (Home, Trends, notifications, avatar), with links to profile, trend analysis, and notifications.
4	xemchitietbaiba	Search page and article detail view — calls timkiem.php to search, and yeuthich.php to save favorite papers.
5	phantichxuhuong	Trend analysis page by topic — calls phantichxuhuong.php, supports following topics and receiving notifications.
6	hosocanhon.html	Account information page — view/edit full name, email, username, email verification status; calls hosocanhon.php, capnhathoso.php, dangxuat.php.
7	yeuthich.html	List of favorited papers, calls yeuthich.php.
8	xemthongbao.htm	Page displaying all notifications, calls thongbao.php.
9	admin.html	System administration page — manages users, topics, API sources, and data synchronization.

1.2.2 Back End



No	Group / File	Description
1	dangnhap.php, dangky.php, dangxuat.php, check_session.ph	Handles user authentication via PHP sessions, with hashed passwords (<code>password_verify</code>).
2	hosocanhon.php, capnhat_hoso.php	Retrieves and updates personal profile information.
3	timkiem.php	Searches for research papers — queries both OpenAlex and Crossref, then merges the results.
4	xuhuongchude.php	Analyzes trending topics — uses Crossref as the primary data source.
5	phantichxuhuong.	Processes trend analysis logic for topics already stored in the database.
6	theo_doi_chude.php yeuthich.php, thongbao.php	Handles topic following, favorite papers, and user notifications.
7	admin_users.php, admin_papers.php admin_topics.php admin_api_source admin_notificati admin_check.php	Administrative functions: managing users, papers, topics, API sources, and notifications.
8	admin_sync.php	Synchronizes paper data from OpenAlex and Semantic Scholar into the database.
9	api_proxy.php	Proxy that calls OpenAlex directly for the dashboard, with file-based caching (10-minute TTL) and rate limiting (5 requests/second).
10	config/database.	Establishes a PDO connection to the MySQL database (<code>sjptts</code>), required by most files in <code>api</code> via <code>require_once</code> .

2 Database Design



Entities Description

2.1 Role Table

Field	Type	Description	PK	FK
roleId	INT	Role ID	Yes	No
roleName	NVARCHAR(100)	Role name	No	No
description	NVARCHAR(255)	Description	No	No

2.2 User Table

Field	Type	Description	PK	FK
userId	INT	User ID	Yes	No
fullName	NVARCHAR(150)	Full name	No	No
email	NVARCHAR(150)	Email	No	No
password	NVARCHAR(255)	Password	No	No
status	NVARCHAR(50)	Status	No	No
createdAt	DATETIME2	Created time	No	No
roleId	INT	Role reference	No	Yes → Role

2.3 Journal Table

Field	Type	Description	PK	FK
journalId	INT	Journal ID	Yes	No
journalName	NVARCHAR(200)	Journal name	No	No
publisher	NVARCHAR(150)	Publisher	No	No
issn	NVARCHAR(20)	ISSN	No	No

2.4 UserFollowJournal (M:N)

Field	Type	Description	PK	FK
userId	INT	User	Yes (Com- posite)	Yes → User
journalId	INT	Journal	Yes (Com- posite)	Yes → Journal

2.5 Topic Table

Field	Type	Description	PK	FK
topicId	INT	Topic ID	Yes	No
topicName	NVARCHAR(150)	Topic name	No	No
description	NVARCHAR(MAX)	Description	No	No
journalId	INT	Journal reference	No	Yes → Journal

2.6 UserFollowTopic (M:N)

Field	Type	Description	PK	FK
userId	INT	User	Yes (Com- posite)	Yes → User
topicId	INT	Topic	Yes (Com- posite)	Yes → Topic

2.7 APISource Table

Field	Type	Description	PK	FK
-------	------	-------------	----	----

sourceId	INT	Source ID	Yes	No
sourceName	NVARCHAR(150)	Source name	No	No
endpointUrl	NVARCHAR(500)	API URL	No	No
syncSchedule	NVARCHAR(100)	Sync schedule	No	No

2.8 Author Table

Field	Type	Description	PK	FK
authorId	INT	Author ID	Yes	No
fullName	NVARCHAR(150)	Full name	No	No
affiliation	NVARCHAR(255)	Affiliation	No	No

2.9 Keyword Table

Field	Type	Description	PK	FK
keywordId	INT	Keyword ID	Yes	No
keywordName	NVARCHAR(100)	Keyword	No	No

2.10 Research Paper Table

Field	Type	Description	PK	FK
articleId	INT	Paper ID	Yes	No
title	NVARCHAR(500)	Title	No	No
abstract	NVARCHAR(MAX)	Abstract	No	No
doi	NVARCHAR(200)	DOI	No	No
publicationYear	INT	Year	No	No
sourceId	INT	API Source	No	Yes → APISource

2.11 PaperTopic (M:N)

Field	Type	Description	PK	FK
articleId	INT	Paper	Yes (Com- posite)	Yes → Re- search- Paper
topicId	INT	Topic	Yes (Com- posite)	Yes → Topic

2.12 PaperKeyword (M:N)

Field	Type	Description	PK	FK
articleId	INT	Paper	Yes (Com- posite)	Yes → Re- search- Paper
keywordId	INT	Keyword	Yes (Com- posite)	Yes → Keyword

2.13 PaperAuthor (M:N)

Field	Type	Description	PK	FK
articleId	INT	Paper	Yes (Com- posite)	Yes → Re- search- Paper
authorId	INT	Author	Yes (Com- posite)	Yes → Author
authorOrder	INT	Author order	No	No

2.14 PublicationTrend Table

Field	Type	Description	PK	FK
trendId	INT	Trend ID	Yes	No
year	INT	Year	No	No
publicationCount	INT	Count	No	No
growthRate	FLOAT	Growth rate	No	No
keywordId	INT	Keyword	No	Yes → Keyword

2.15 Report Table

Field	Type	Description	PK	FK
reportId	INT	Report ID	Yes	No
title	NVARCHAR(300)	Title	No	No
keyword	NVARCHAR(200)	Keyword	No	No
yearFrom	INT	From year	No	No
yearTo	INT	To year	No	No
generatedAt	DATETIME2	Created time	No	No
userId	INT	User	No	Yes → User

2.16 Notification Table

Field	Type	Description	PK	FK
notificationId	INT	Notification ID	Yes	No
title	NVARCHAR(300)	Title	No	No
content	NVARCHAR(MAX)	Content	No	No
isRead	BIT	Read status	No	No

createdAt	DATETIME2	Created time	No	No
userId	INT	User	No	Yes → User

2.17 NotificationPaper (M:N)

Field	Type	Description	PK	FK
notificationId	INT	Notification	Yes (Composite)	Yes → Notification
articleId	INT	Paper	Yes (Composite)	Yes → Research-Paper

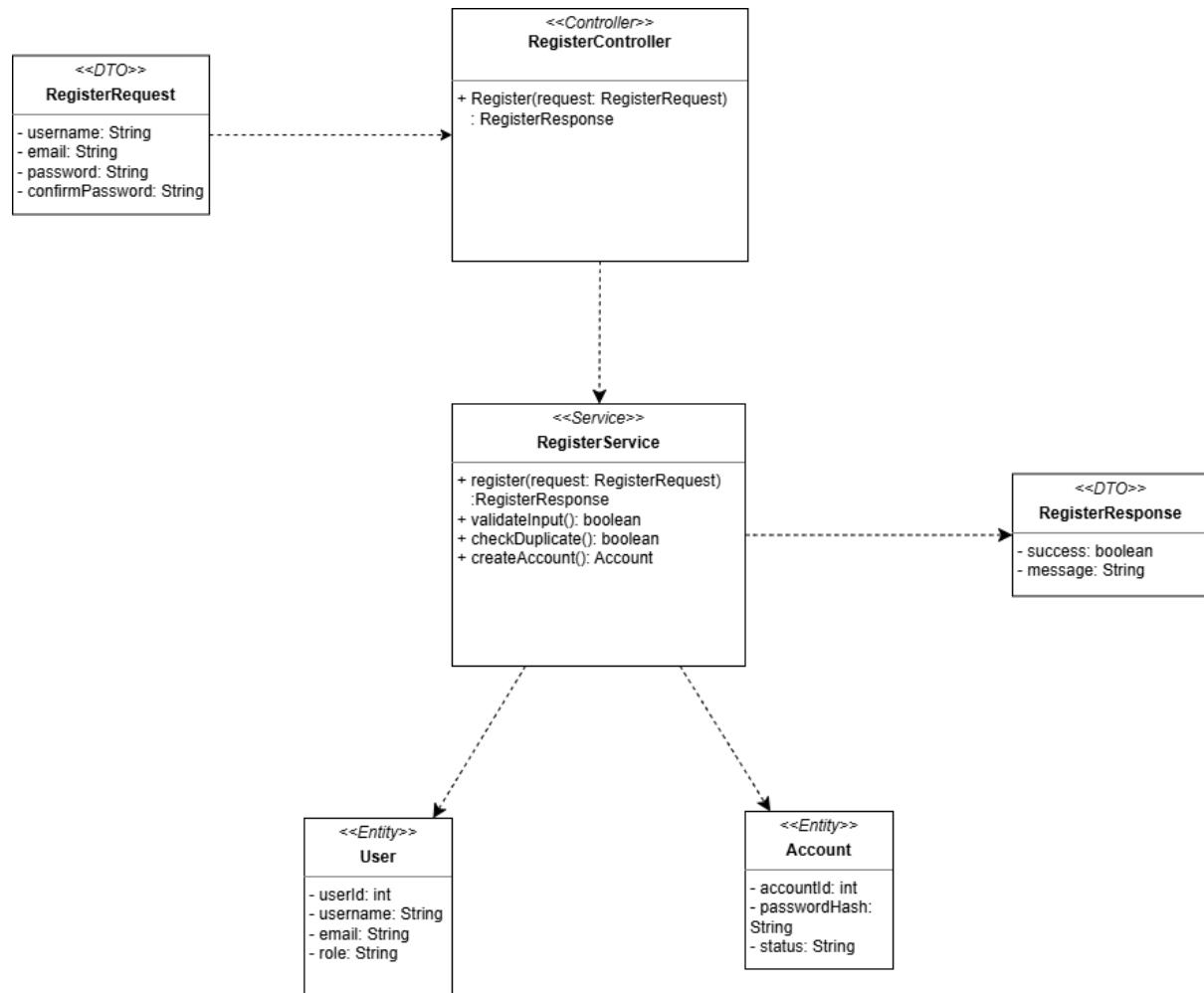
2.18 Bookmark Table

Field	Type	Description	PK	FK
bookmarkId	INT	Bookmark ID	Yes	No
bookmarkedAt	DATETIME2	Time saved	No	No
userId	INT	User	No	Yes → User
articleId	INT	Paper	No	Yes → Research-Paper
topicId	INT	Topic	No	Yes → Topic

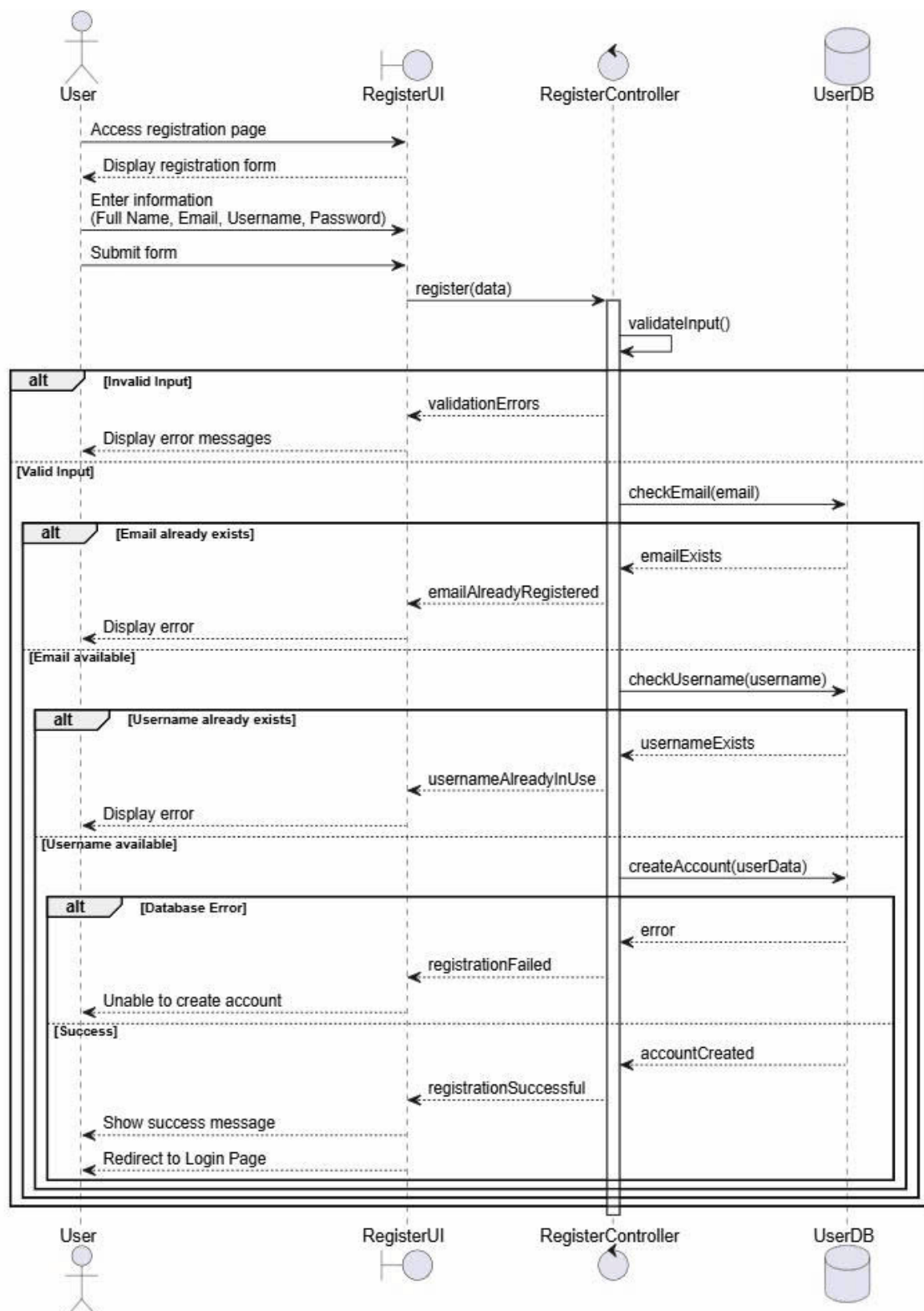
3 Detail Design

3.1 Register

3.1.1 Class Diagram

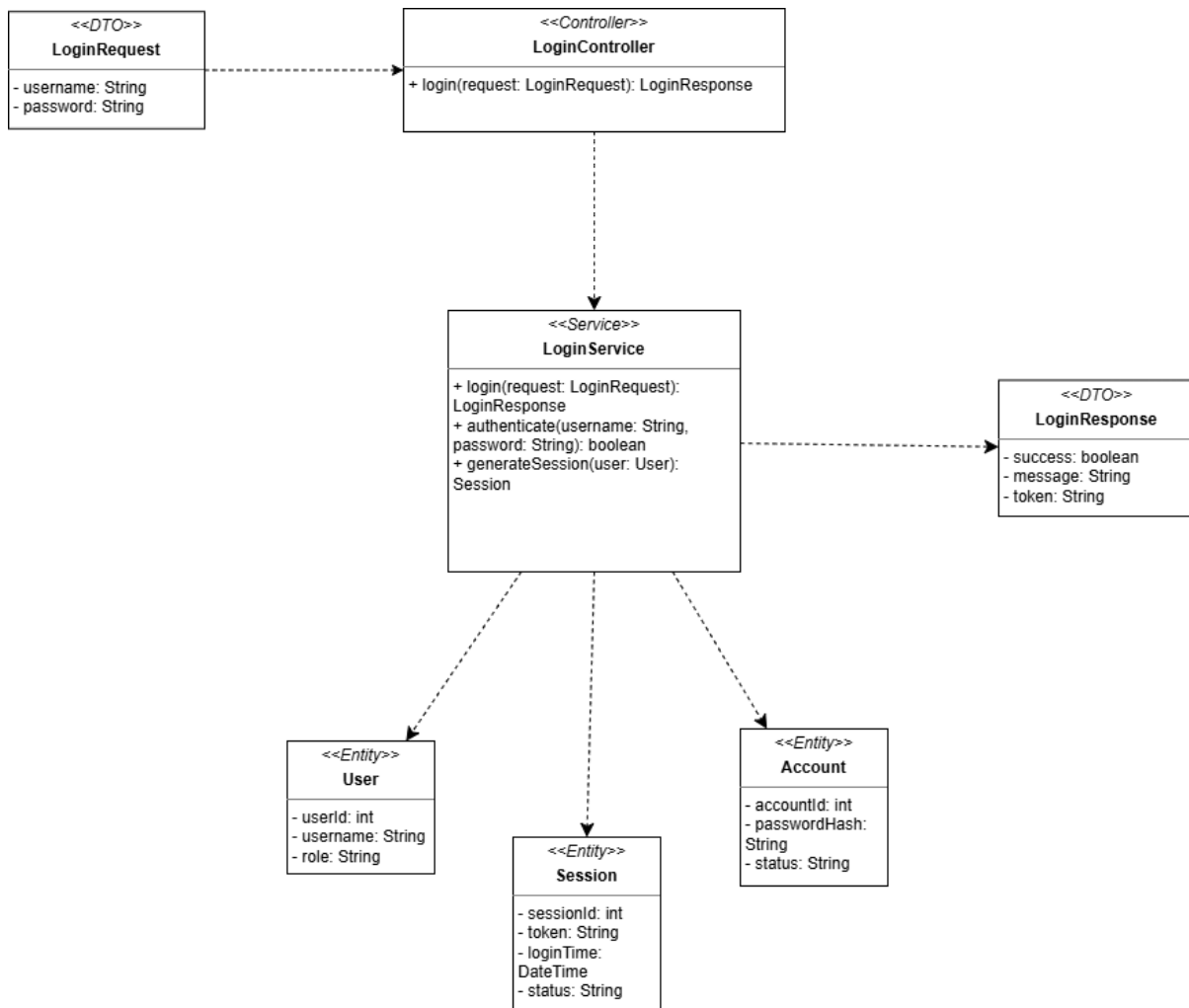


3.1.2 Sequence Diagram

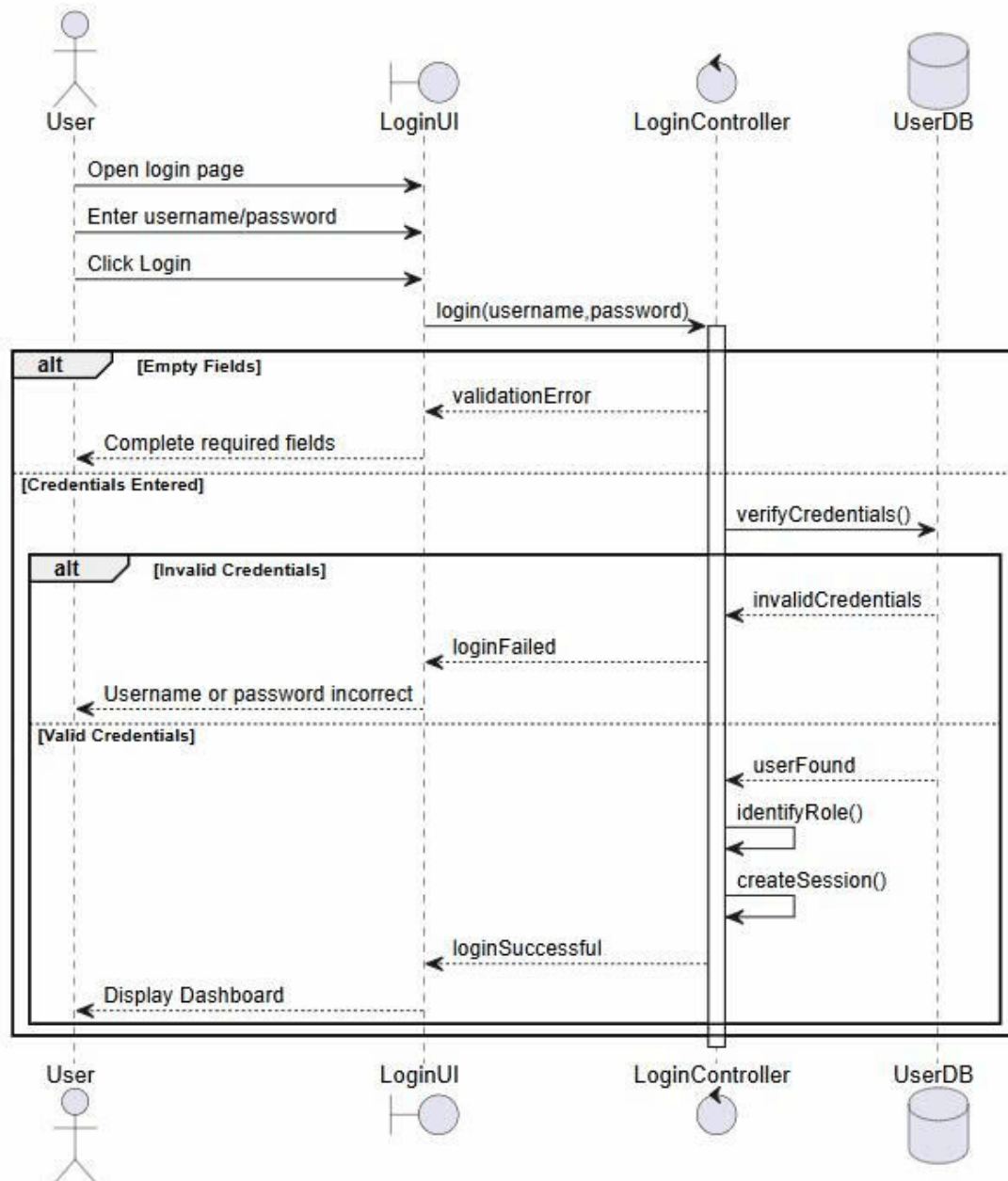


3.2 Login

3.2.1 Class Diagram

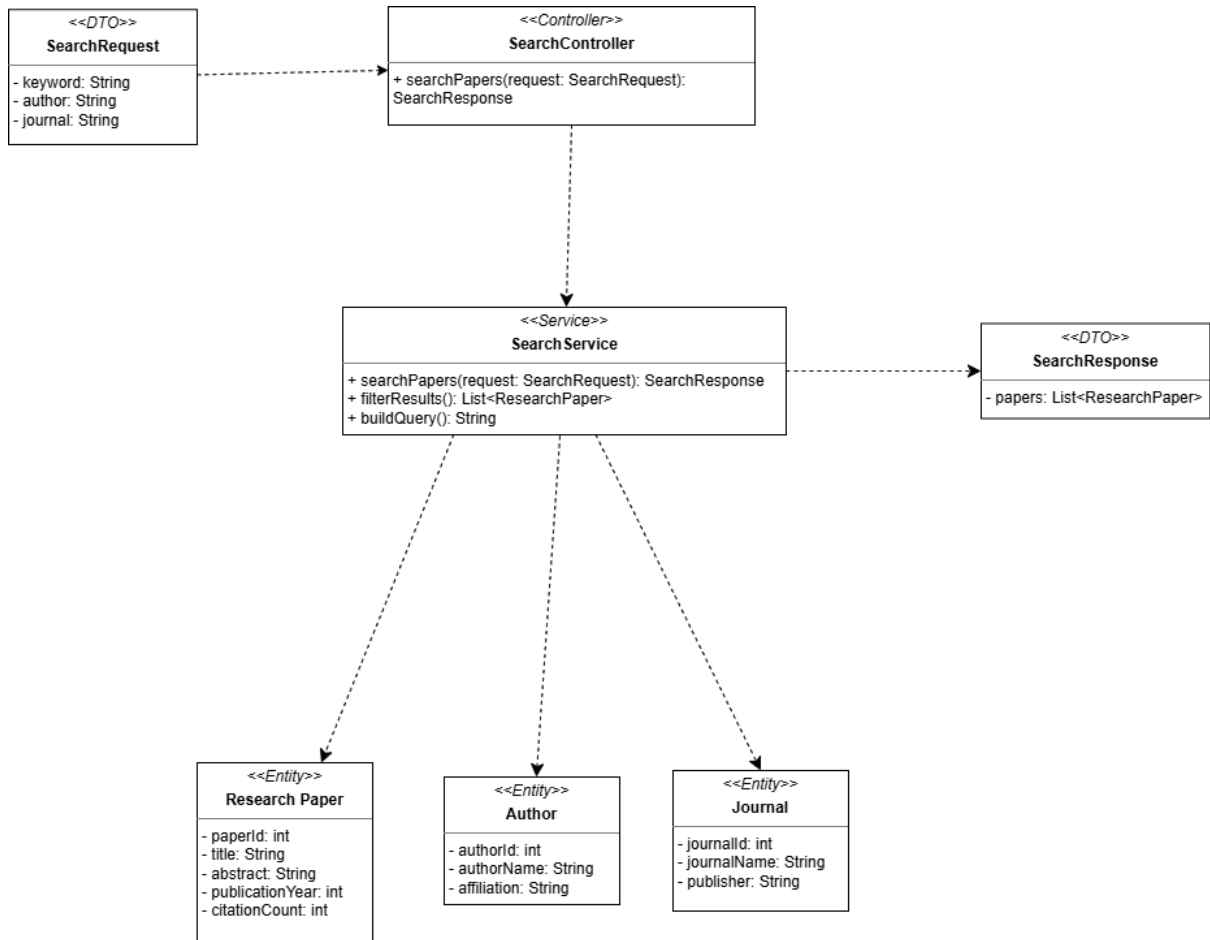


3.2.2 Sequence Diagram



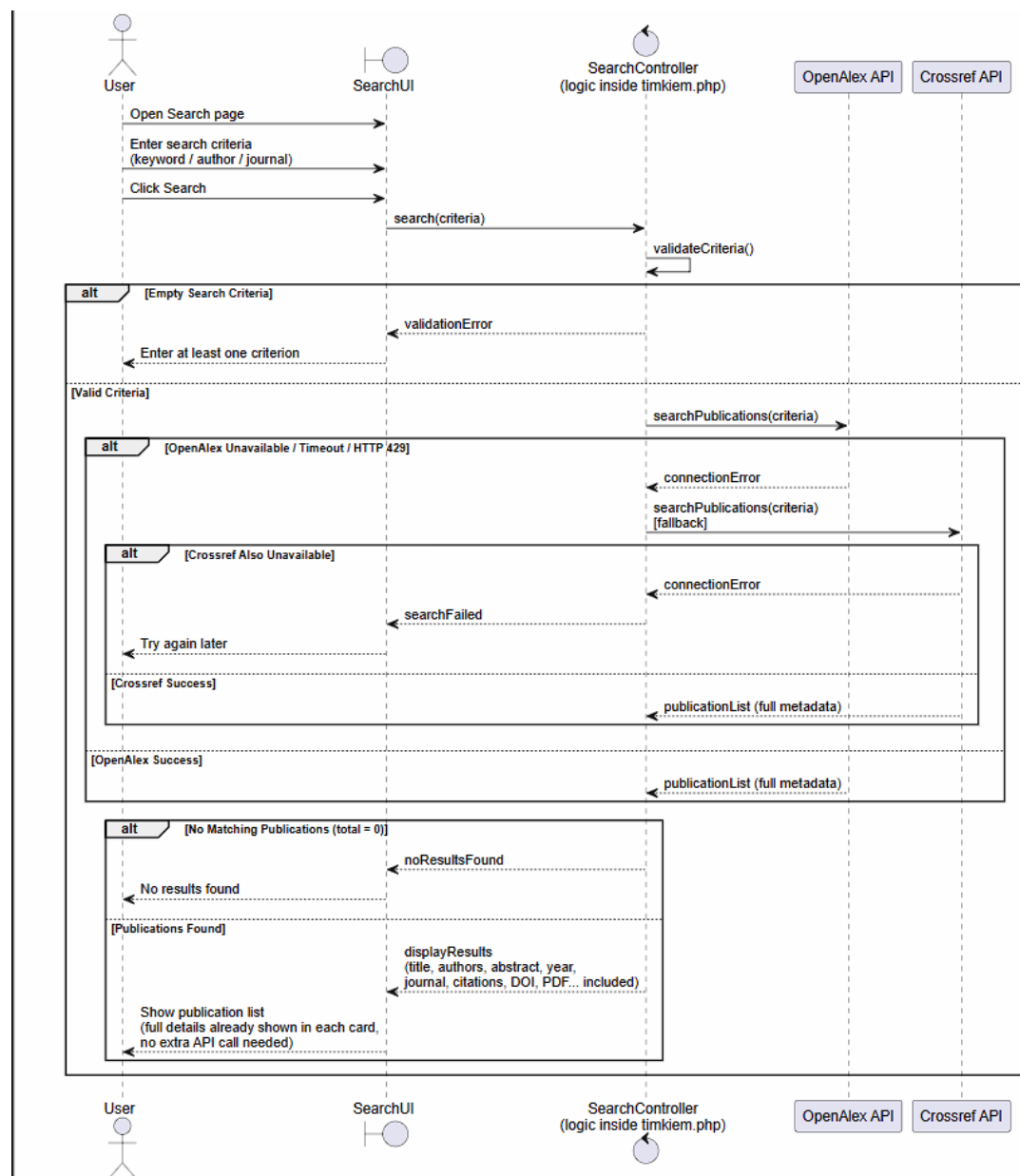
3.3 Search Research Papers

3.3.1 Class Diagram



Hình 1: Class Diagram: Search Research Papers

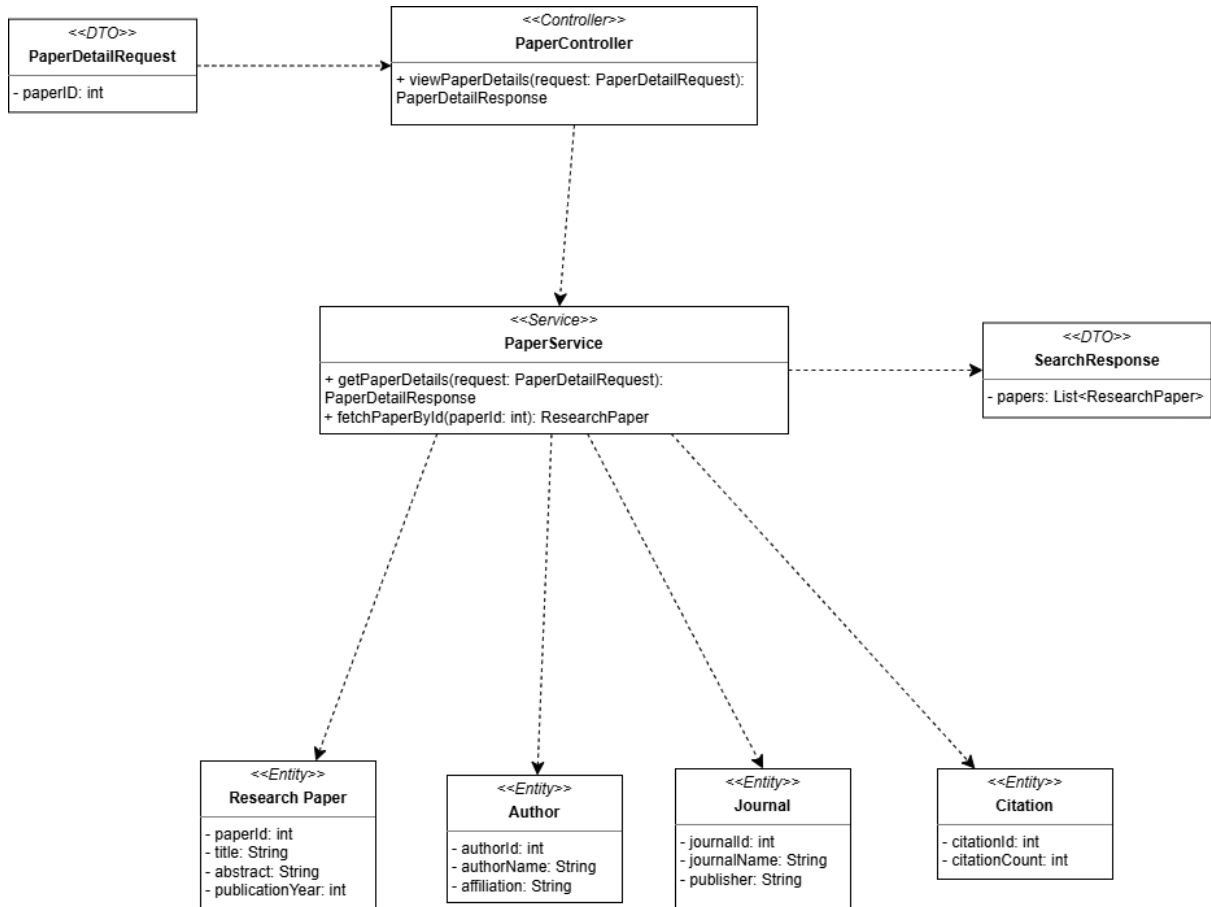
3.3.2 Sequence Diagram



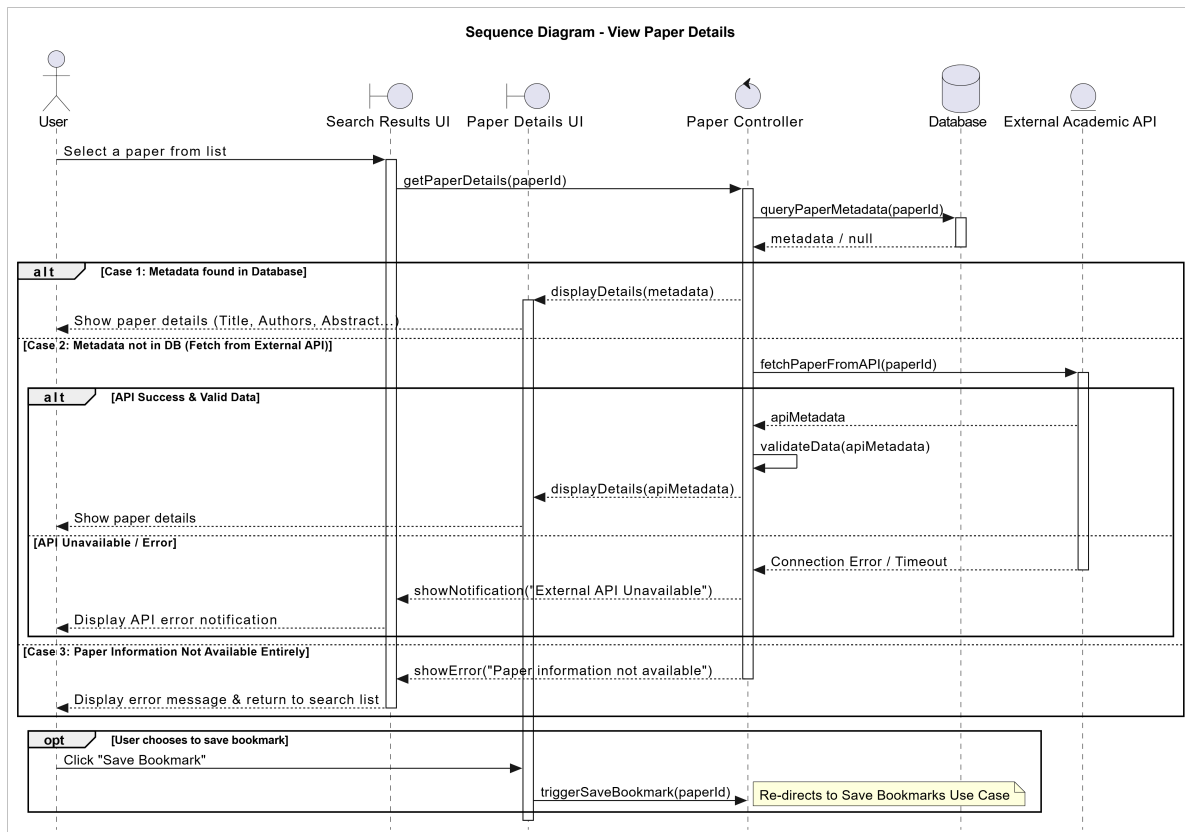
Hình 2: Sequence Diagram

3.4 View Paper Details

3.4.1 Class Diagram

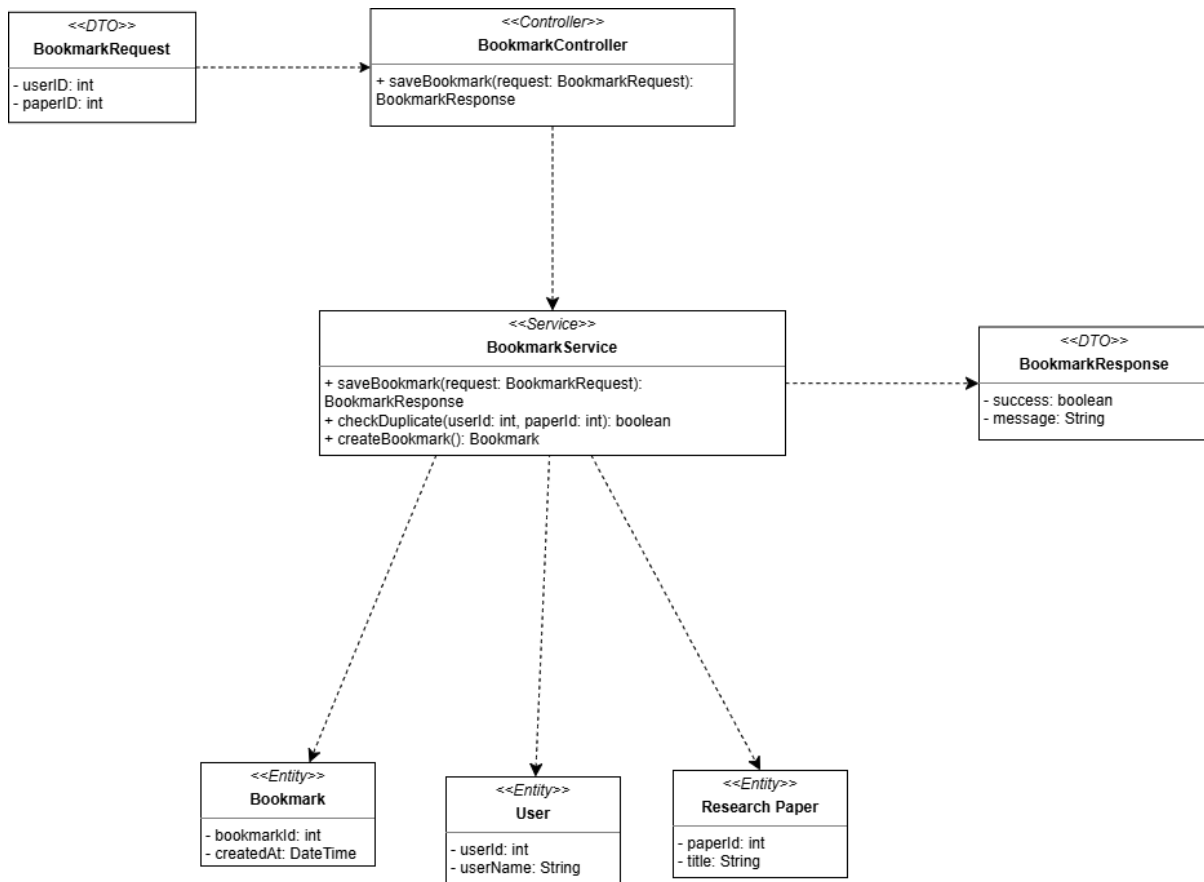


3.4.2 Sequence Diagram

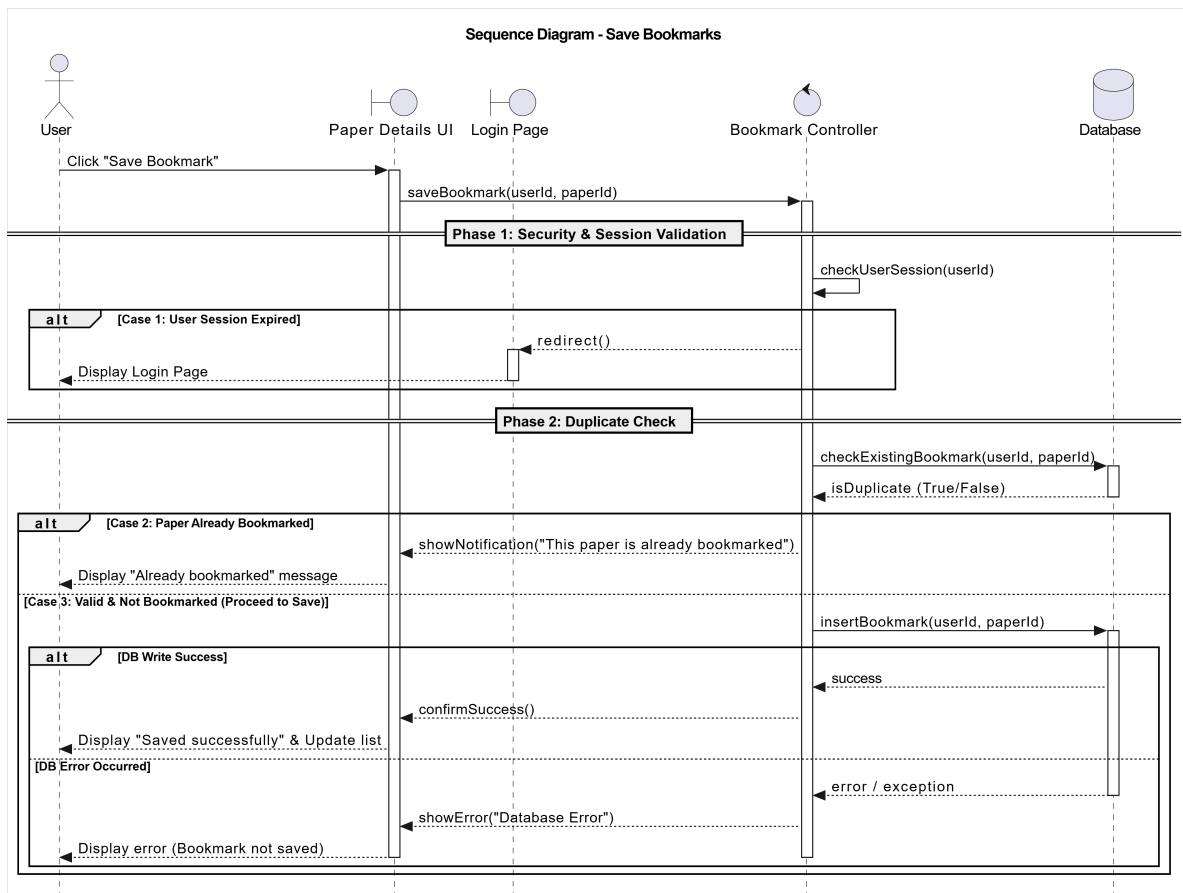


3.5 Save Bookmarks

3.5.1 Class Diagram

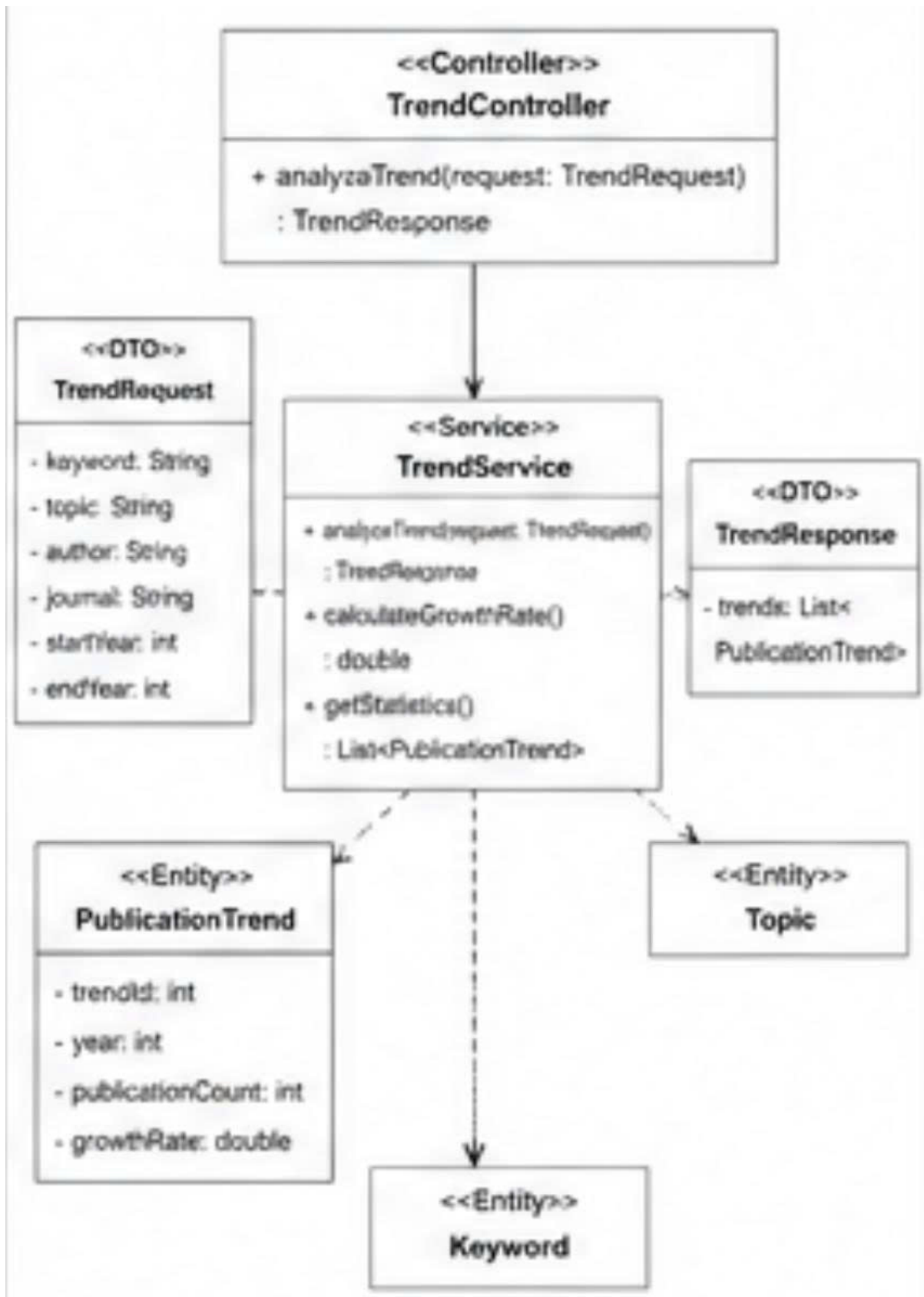


3.5.2 Sequence Diagram



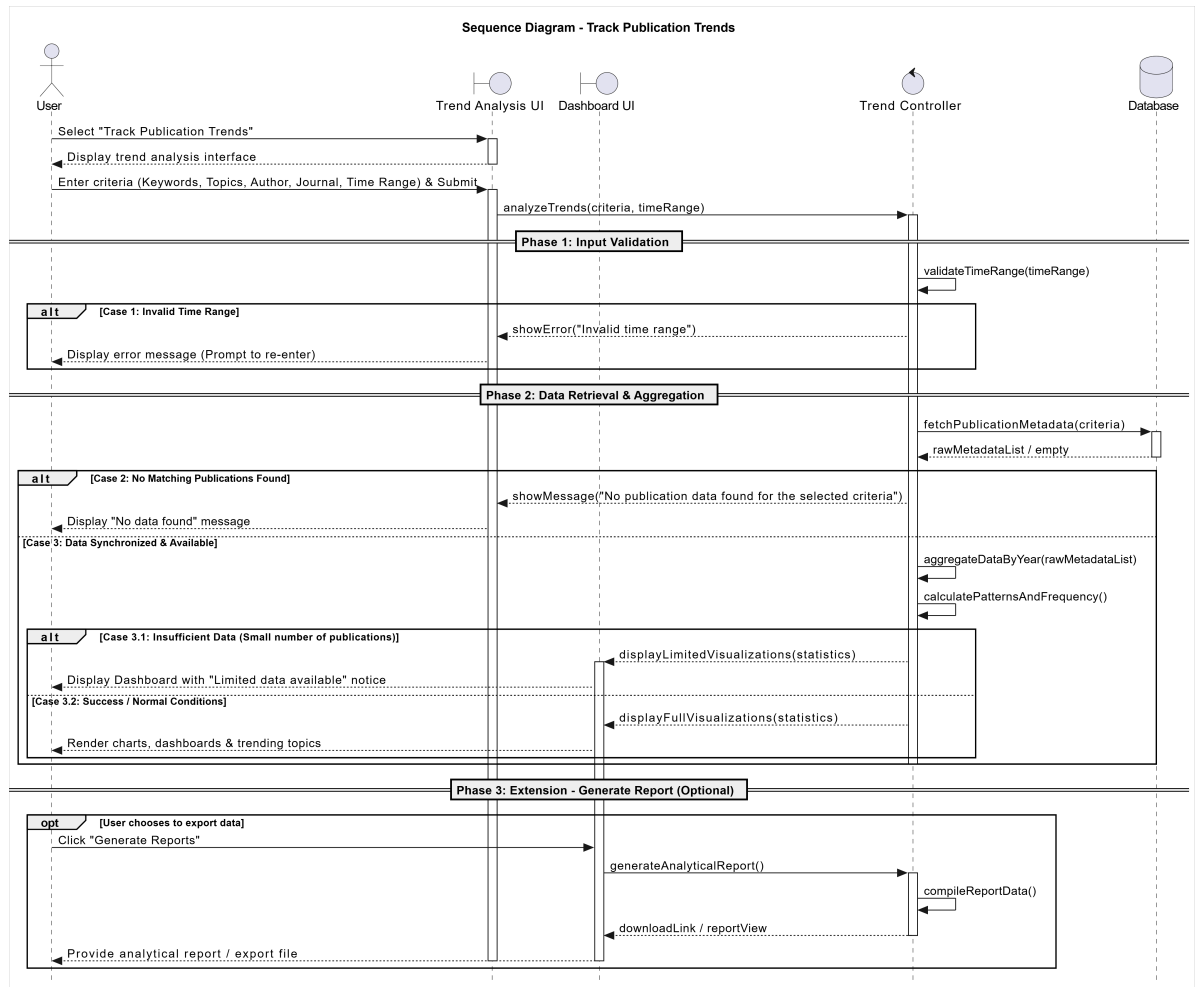
3.6 Track Publication Trends

3.6.1 Class Diagram



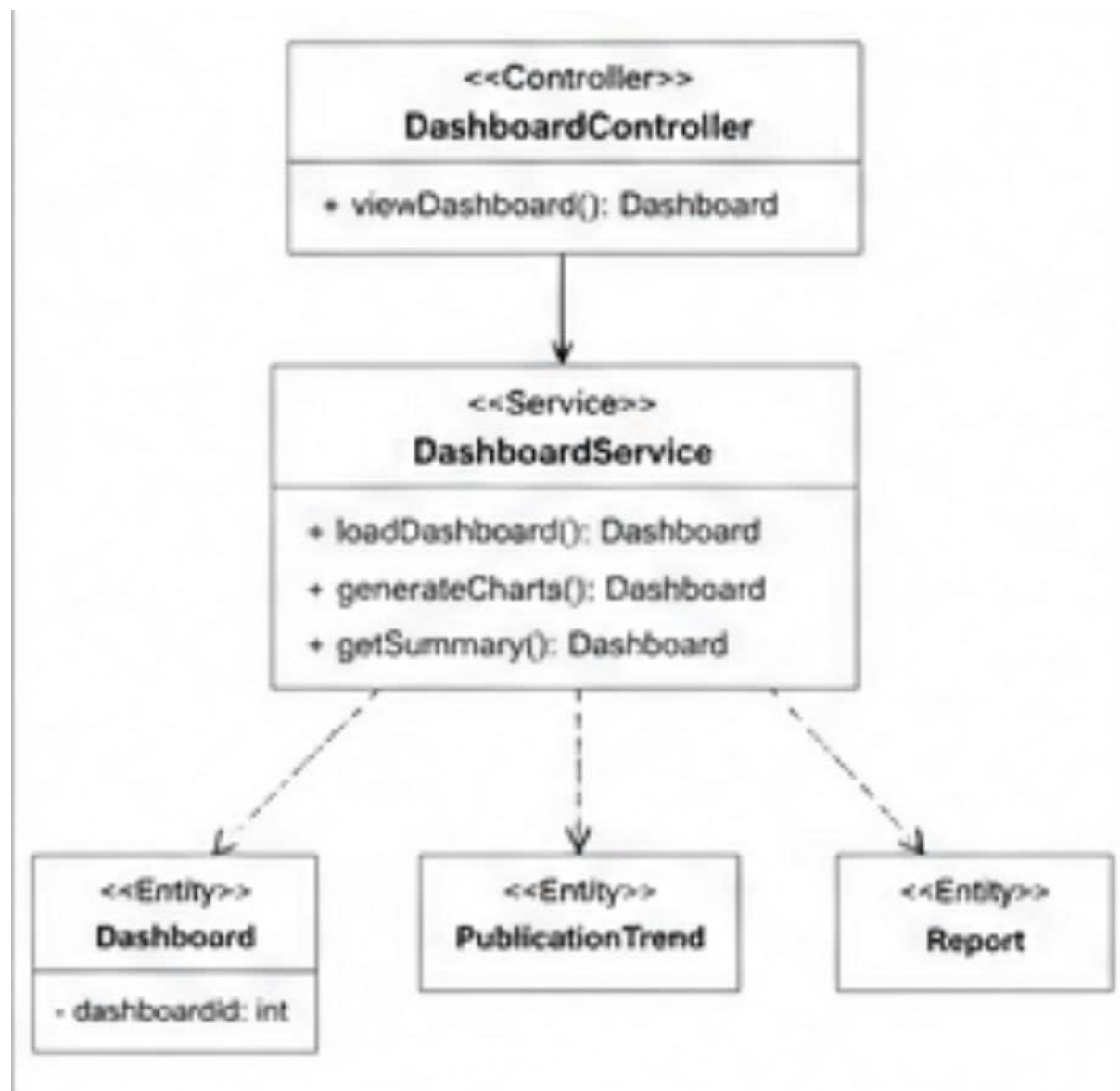
Hình 3: Class Diagram: Track Publication Trends

3.6.2 Sequence Diagram



3.7 View Dashboard

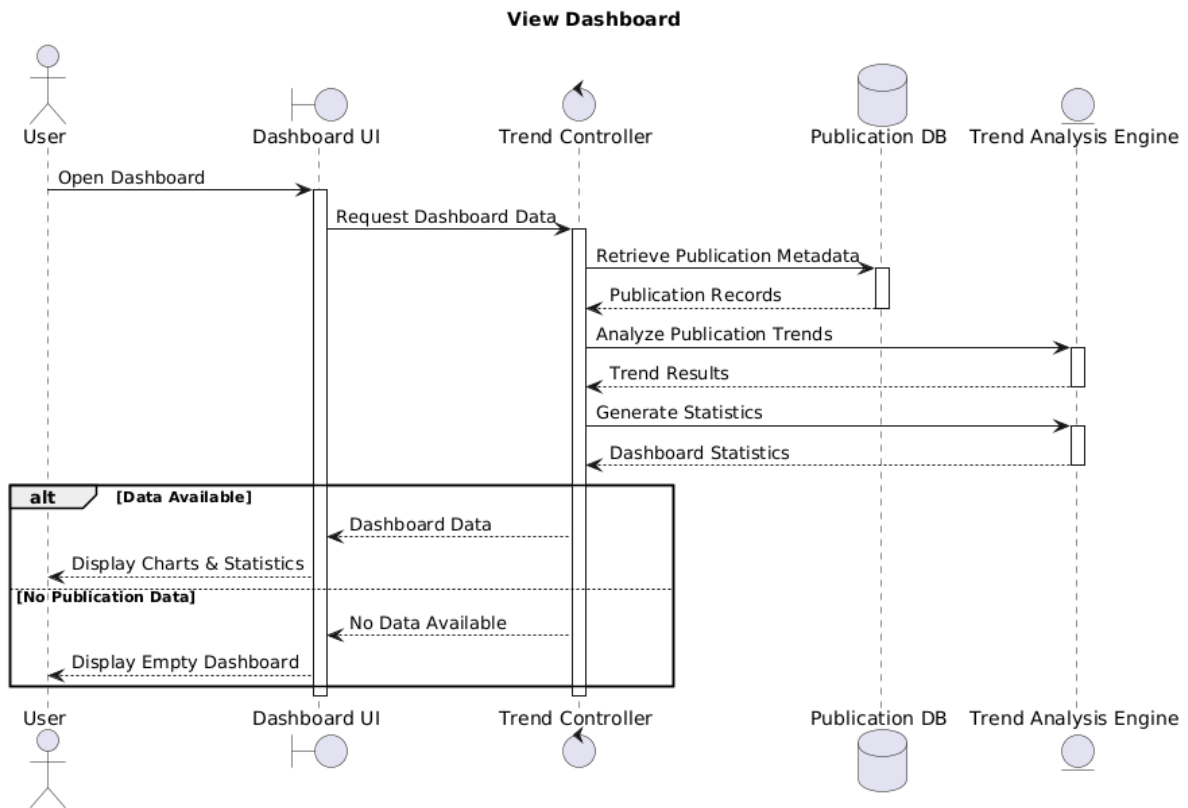
3.7.1 Class Diagram



Hình 4: Class Diagram: View Dashboard

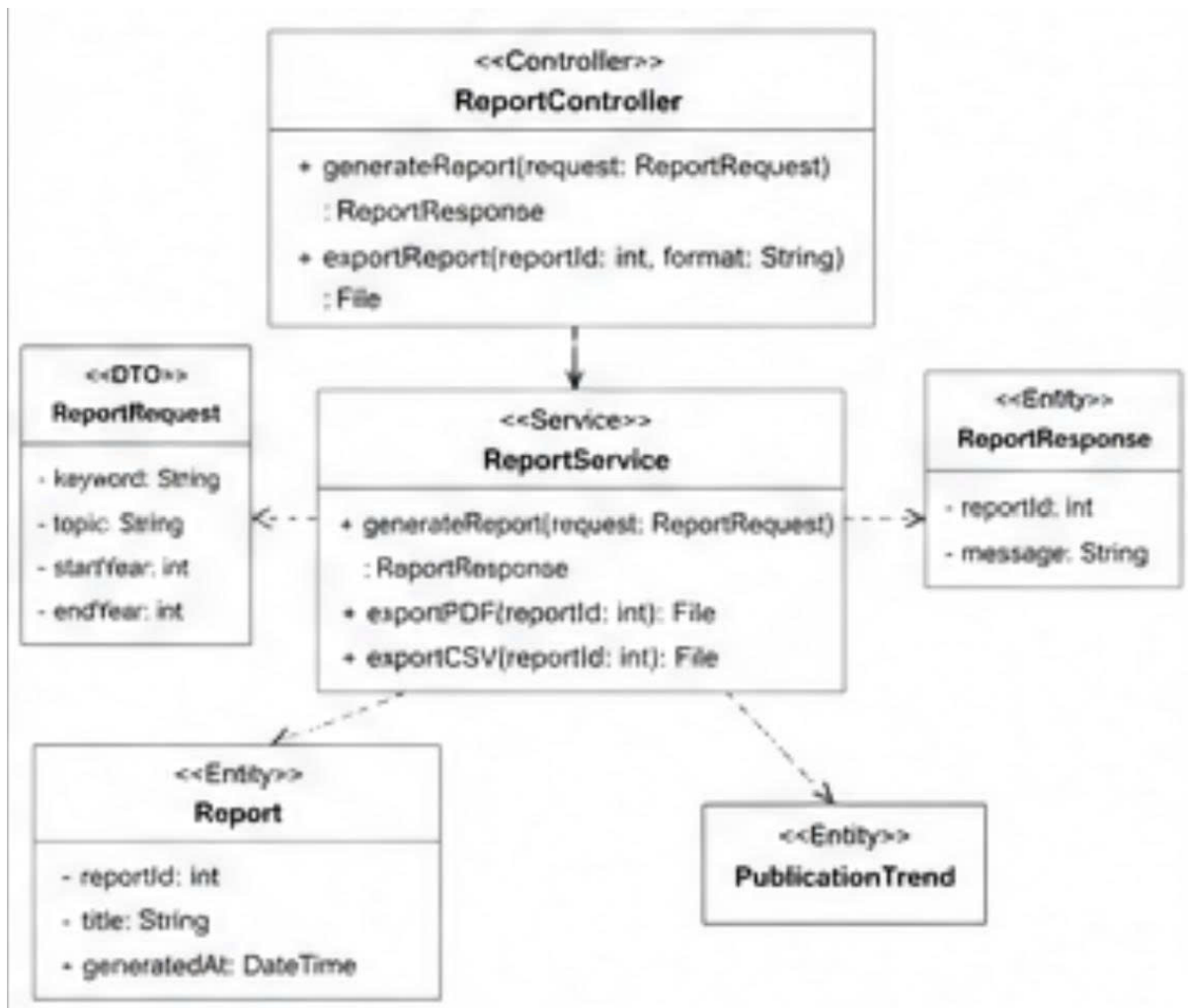
3.8 View Dashboard

3.8.1 Sequence Diagram



3.9 Generate Reports

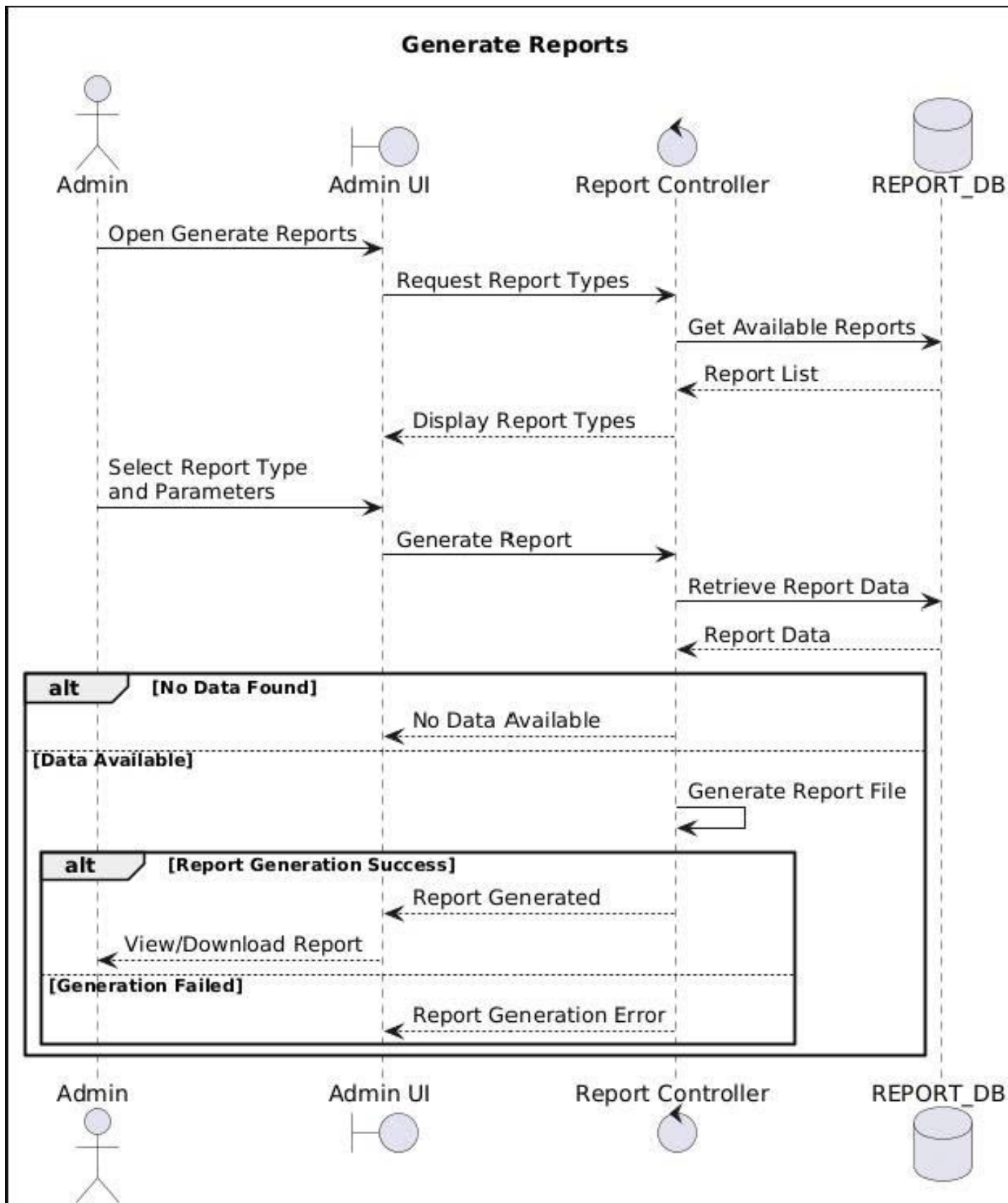
3.9.1 Class Diagram



Hình 5: Class Diagram - Generate Reports

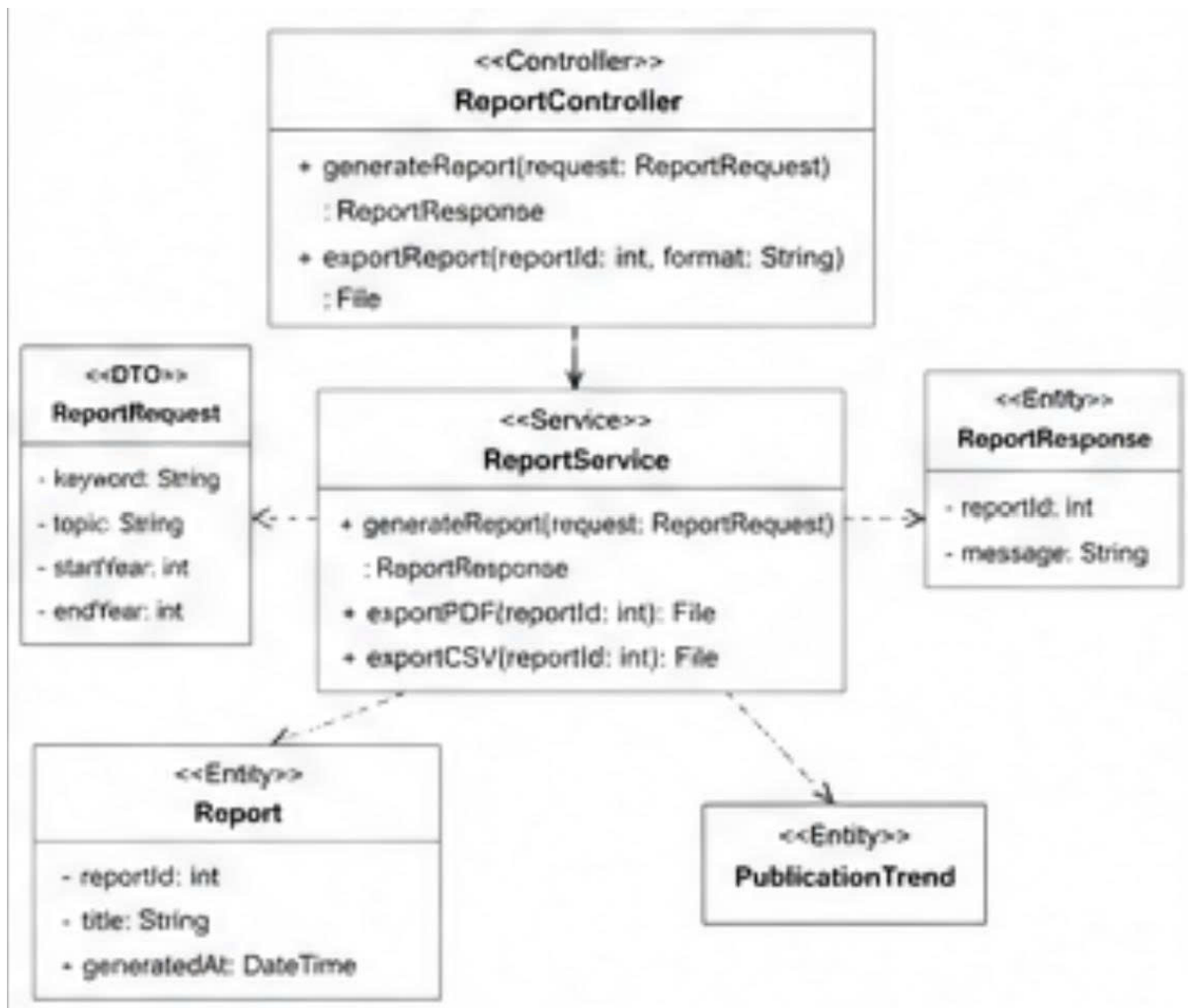
3.10 Generate Reports

3.10.1 Sequence Diagram



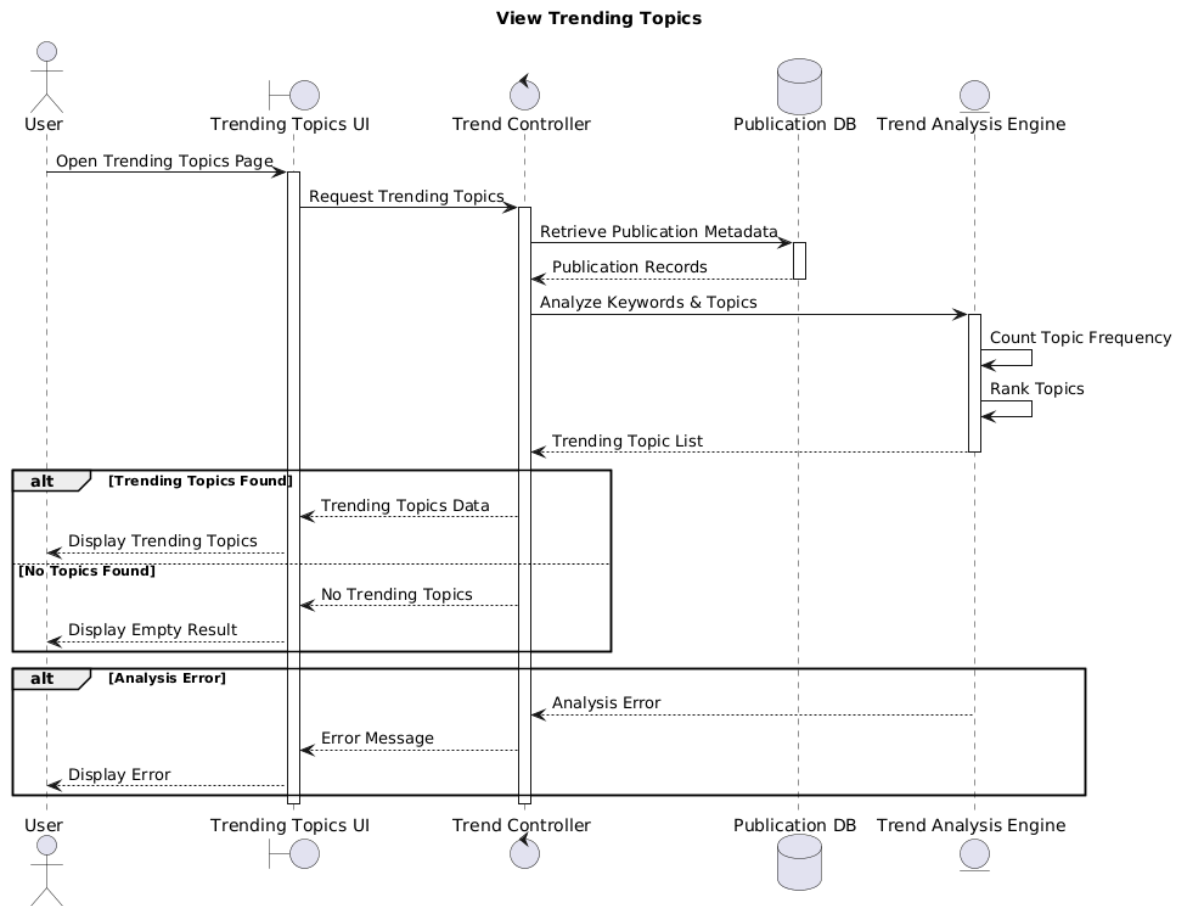
3.11 View Trending Topics

3.11.1 Class Diagram



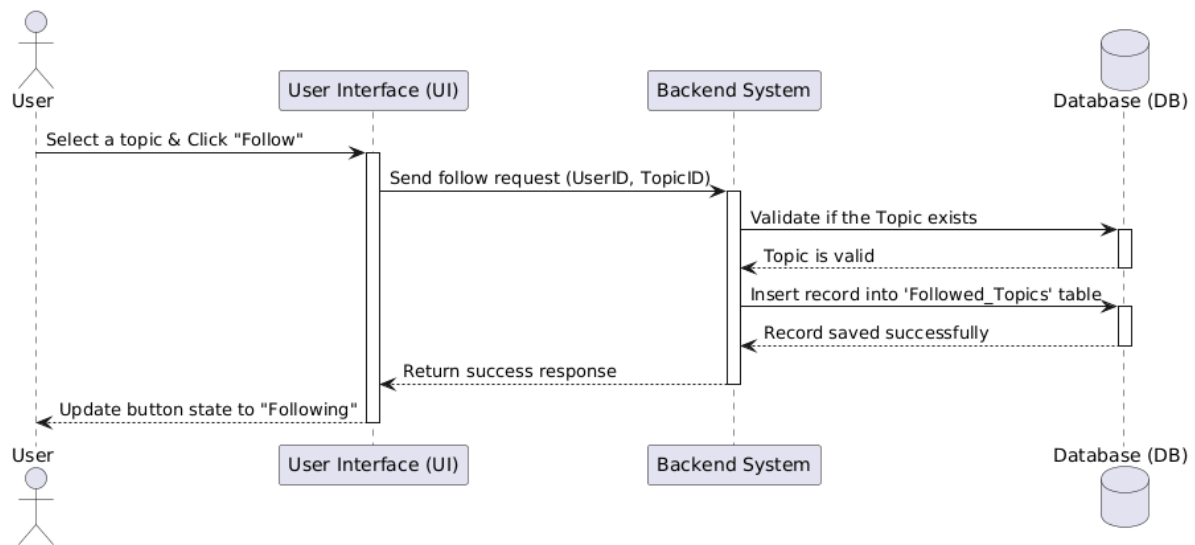
3.12 View Trending Topics

3.12.1 Sequence Diagram



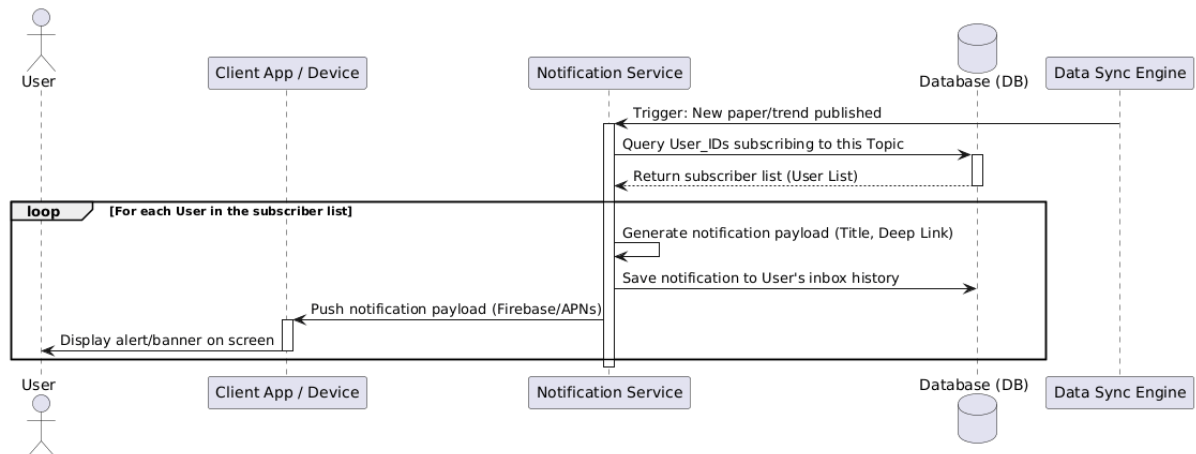
3.13 Follow Topics

3.13.1 Sequence Diagram



3.14 Receive Notifications

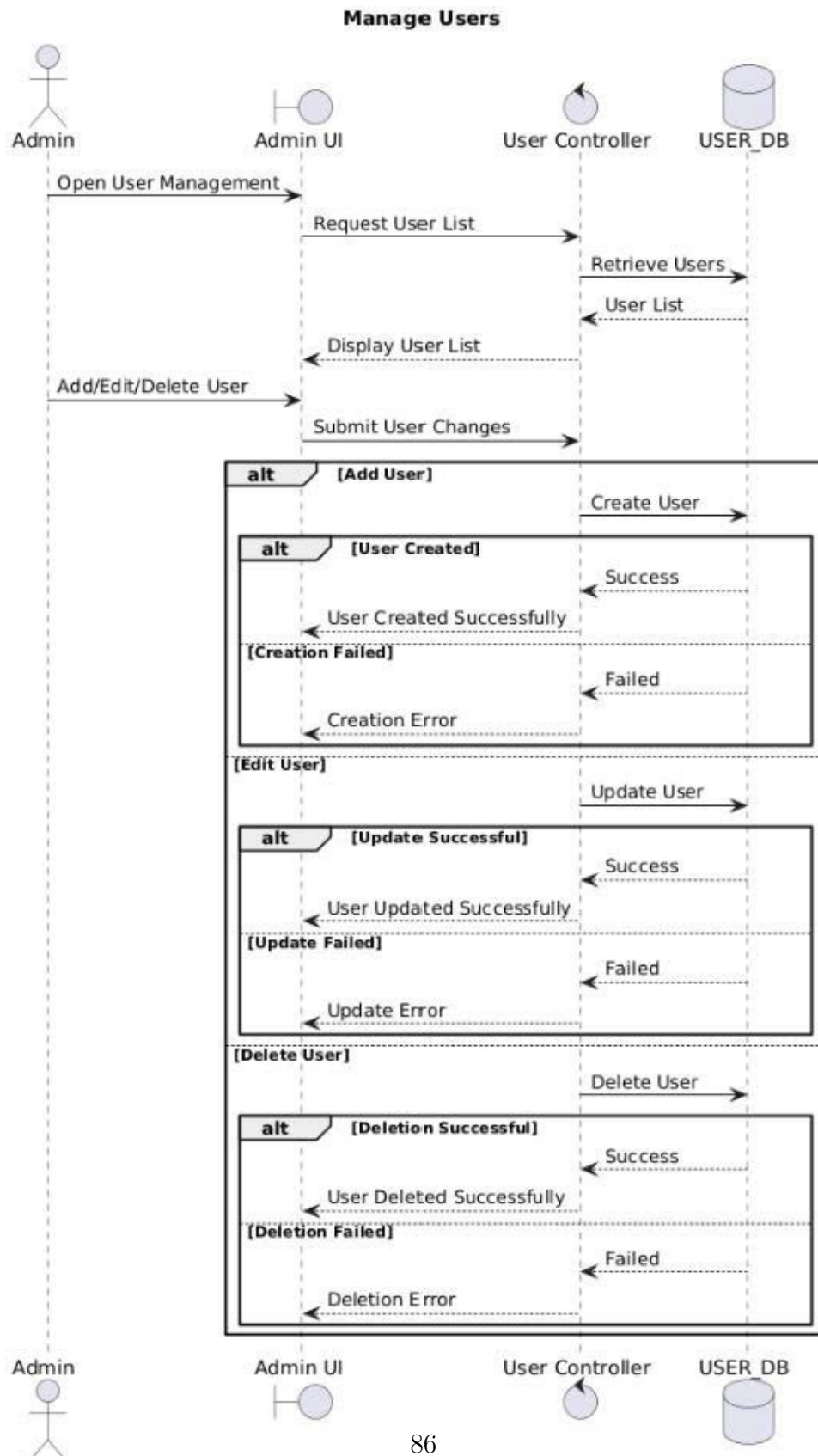
3.14.1 Sequence Diagram



3.15 Manage System

3.16 Manage User

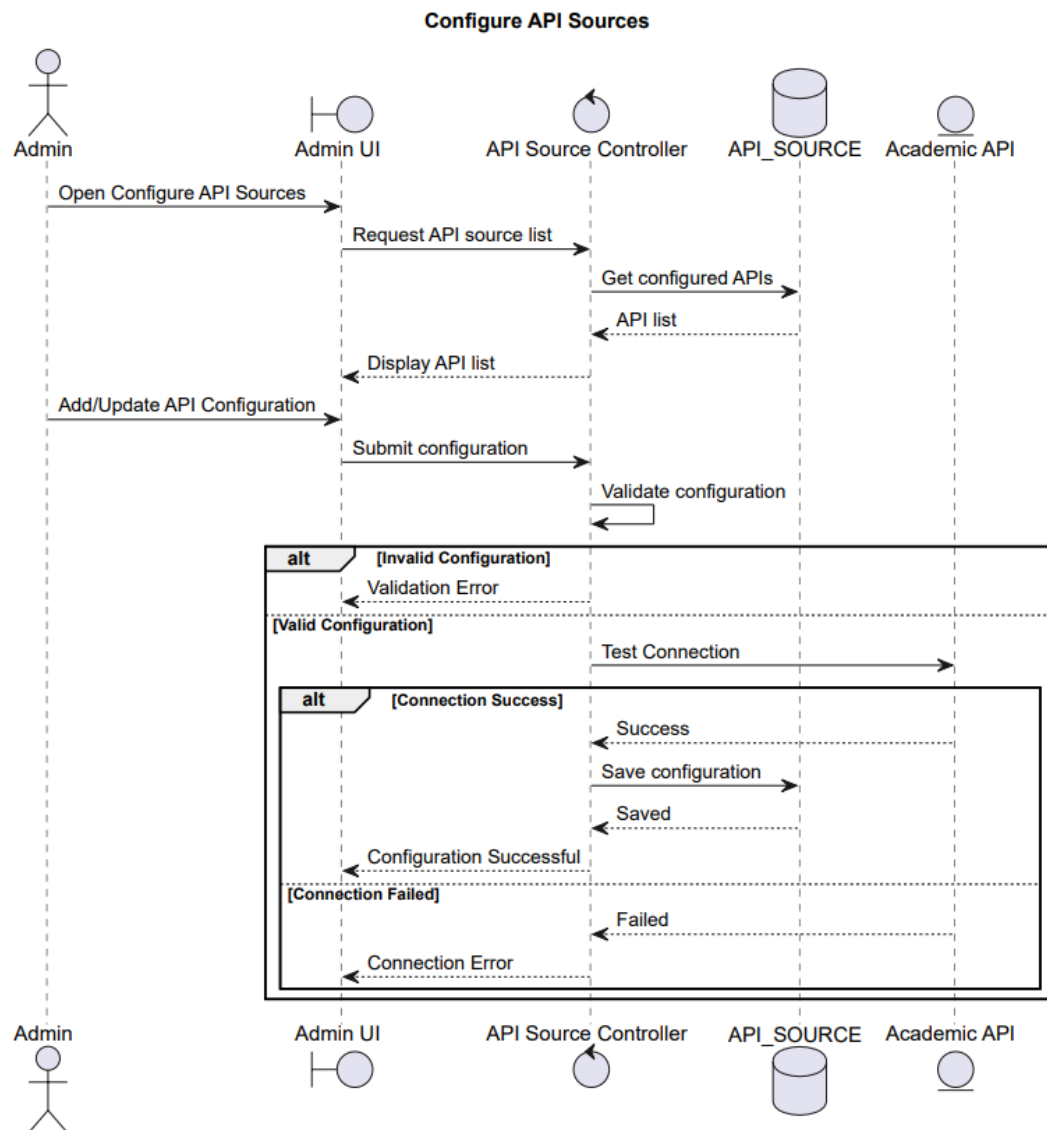
3.16.1 Sequence Diagram



3.17 Configure API Sources

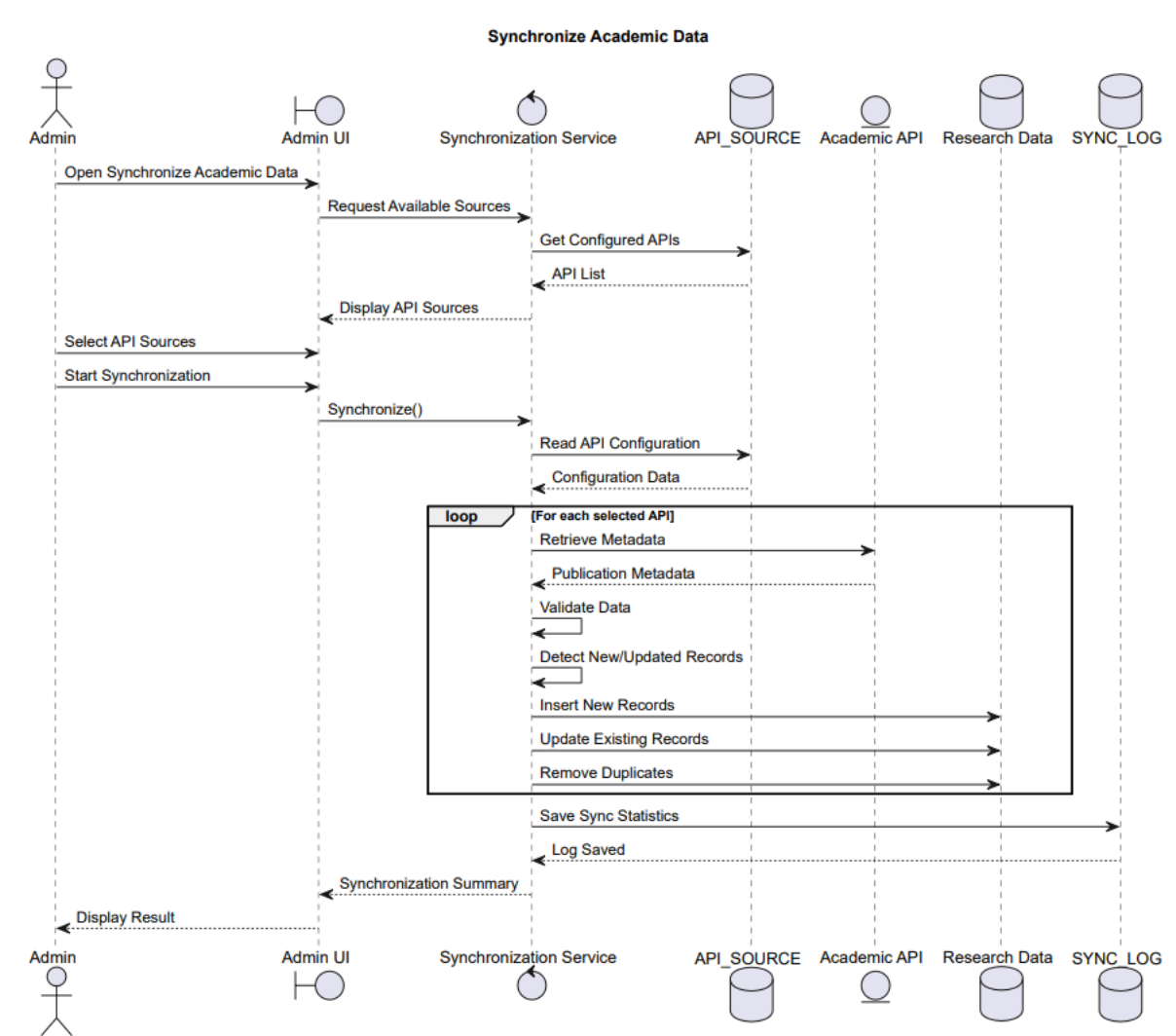
3.18 Configure API Sources

3.18.1 Sequence Diagram



3.19 Sync Academic Data

3.19.1 Sequence Diagram



V. SOFTWARE TESTING DOCUMENTATION

1 Testing Scope

The testing scope covers all major functionalities of the Scientific Journal Publication Trend Tracking System. The following modules and features are included within the scope of testing:

- User Registration
- User Login and Authentication
- Search Scientific Publications
- View Publication Details
- Save and Manage Bookmarks
- Track Publication Trends
- View Statistical Dashboard
- Generate Reports (PDF/CSV)
- View Trending Topics
- Follow Research Topics
- Receive Notifications

The following features are excluded from the testing scope:

- Third-party services and external systems that are outside the project scope.
- Real-time publication trend prediction using Artificial Intelligence (AI) or Machine Learning (ML) techniques.
- Performance testing under extremely large-scale environments involving more than 10,000 concurrent users.

2 Test Strategy

2.1 Testing Types

- **Functional Testing:** Verify that all system functions operate according to the specified requirements.
- **Integration Testing:** Verify communication and data exchange between the frontend, backend, database, and external APIs.

- **System Testing:** Validate complete business workflows and end-to-end system functionality.
- **Performance Testing:** Evaluate system response time and behavior under different workloads.

2.2 Testing Levels

- **Unit Testing:** Verify the correctness of individual modules using white-box testing techniques.
- **Integration Testing:** Verify interactions among integrated components and services.
- **System Testing:** Validate complete business processes using black-box testing techniques.
- **Performance Testing:** Observe system behavior and collect performance metrics under different workloads.

2.3 Supporting Tools

Tool	Purpose
Visual Studio Code	Application development and source code editing
MySQL	Database validation and data storage testing
phpMyAdmin	Database administration and query verification
Google Chrome	User interface and functional testing
Postman	API testing and response validation
GitHub	Source code version control and collaboration

3 Test Plan

Item	Description
Environment	Windows 11, WampServer, MySQL
Browser	Google Chrome
Database	MySQL
External APIs	OpenAlex, CrossRef
Test Period	During system development and before deployment

3.1 Entry Criteria

- Source code has been completed.
- Database configuration has been completed.
- External APIs are accessible.

3.2 Exit Criteria

- All major functionalities have been tested.
- No critical defects remain unresolved.

4 Test Case

Module	Number of Test Cases
User Registration	5
User Login	3
Publication Search	4
Bookmarks	3
Follow Research Topics	3
Notifications	2
Publication Trend Analysis	2
Total	22

4.1 User Registration

Field	Description
Module	User Registration
Test Case ID	TC_REG_01
Test Case Name	Valid Registration
Objective	Verify that a user can successfully create a new account.
Preconditions	Email and username do not exist in the database.
Test Data	Full Name: Group Six Username: Group6 Email: group6@gmail.com Password: Group654321
Test Steps	1. Open Registration page. 2. Enter all required information. 3. Click Register.
Expected Result	Account is created successfully and saved in the database.
Actual Result	Account created successfully.
Status	Pass

Field	Description
Module	User Registration
Test Case ID	TC_REG_02
Test Case Name	Empty Required Fields
Objective	Verify validation for mandatory fields.
Preconditions	None
Test Data	All fields are left blank.
Test Steps	1. Open Registration page. 2. Leave all fields empty. 3. Click Register.
Expected Result	System displays a validation message requiring all mandatory fields to be filled.
Actual Result	Validation message displayed correctly.
Status	Pass

Field	Description
Module	User Registration
Test Case ID	TC_REG_03
Test Case Name	Invalid Email Format
Objective	Verify email format validation.
Preconditions	None
Test Data	Email: group6123
Test Steps	1. Open Registration page. 2. Enter an invalid email address. 3. Click Register.
Expected Result	System displays "Invalid email address".
Actual Result	Validation message displayed correctly.
Status	Pass

Field	Description
Module	User Registration
Test Case ID	TC_REG_04
Test Case Name	Weak Password
Objective	Verify password strength validation.
Preconditions	None
Test Data	Password: 123456
Test Steps	1. Open Registration page. 2. Enter a weak password. 3. Click Register.
Expected Result	System displays a password strength validation message.
Actual Result	Validation message displayed correctly.
Status	Pass

Field	Description
Module	User Registration
Test Case ID	TC_REG_05
Test Case Name	Existing Email
Objective	Verify duplicate email validation.
Preconditions	The email already exists in the database.
Test Data	Email: group6@gmail.com
Test Steps	1. Open Registration page. 2. Enter an existing email. 3. Click Register.
Expected Result	System displays "Email already registered".
Actual Result	Validation message displayed correctly.
Status	Pass

4.2 User Login

Field	Description
Module	User Login
Test Case ID	TC_LOG_01
Test Case Name	Successful Login
Objective	Verify that users can log in with valid credentials.
Preconditions	User account exists and is active.
Test Data	Username: Group6 Password: Group654321
Test Steps	1. Open Login page. 2. Enter valid username and password. 3. Click Login.
Expected Result	User is redirected to the dashboard successfully.
Actual Result	User logged in successfully.
Status	Pass

Field	Description
Module	User Login
Test Case ID	TC_LOG_02
Test Case Name	Invalid Login Credentials
Objective	Verify system behavior when username or password is incorrect.
Preconditions	User login page is accessible.
Test Data	Scenario 1: Username: Group6 Password: WrongPassword123 Scenario 2: Username: UnknownUser Password: Group654321
Test Steps	1. Open Login page. 2. Enter invalid credentials. 3. Click Login.
Expected Result	System displays “Invalid username or password”.
Actual Result	Error message displayed correctly.
Status	Pass

Field	Description
Module	User Login
Test Case ID	TC_LOG_03
Test Case Name	Empty Credentials
Objective	Verify validation for empty login fields.
Preconditions	None.
Test Data	Username and Password are empty.
Test Steps	1. Open Login page. 2. Leave all fields blank. 3. Click Login.
Expected Result	System displays validation messages for required fields.
Actual Result	Validation messages displayed correctly.
Status	Pass

4.3 Search Scientific Publication

Field	Description
Module	Search Publications
Test Case ID	TC_SEARCH_01
Test Case Name	Search by Valid Keyword
Objective	Verify that users can search publications using a valid keyword.
Preconditions	Publication data exists in the database.
Test Data	Keyword: Machine Learning
Test Steps	1. Enter keyword "Machine Learning". 2. Click Search.
Expected Result	A list of relevant publications is displayed.
Actual Result	Relevant publications displayed successfully.
Status	Pass

Field	Description
Module	Search Publications
Test Case ID	TC_SEARCH_02
Test Case Name	Search with No Results
Objective	Verify system behavior when no publication matches the search keyword.
Preconditions	System is available.
Test Data	Keyword: abcdeftopic123
Test Steps	1. Enter keyword "abcdeftopic123". 2. Click Search.
Expected Result	System displays "No publications found".
Actual Result	No results displayed correctly.
Status	Pass

Field	Description
Module	Search Publications
Test Case ID	TC_SEARCH_03
Test Case Name	Filter by Publication Year
Objective	Verify filtering publications by publication year.
Preconditions	Publications from multiple years exist.
Test Data	Year: 2024
Test Steps	1. Open Search page. 2. Select publication year = 2024. 3. Click Apply Filter.
Expected Result	Only publications published in 2024 are displayed.
Actual Result	Publications filtered correctly by year.
Status	Pass

Field	Description
Module	Search Publications
Test Case ID	TC_SEARCH_04
Test Case Name	Filter Open Access Publications
Objective	Verify filtering publications with Open Access status.
Preconditions	The database contains both Open Access and non-Open Access publications.
Test Data	Open Access = True
Test Steps	1. Open Search page. 2. Enable Open Access filter. 3. Click Apply Filter.
Expected Result	Only Open Access publications are displayed.
Actual Result	Open Access publications displayed correctly.
Status	Pass

4.4 Manage Bookmarks

Field	Description
Module	Bookmark (Favorites)
Test Case ID	TC_BM_01
Test Case Name	Add Publication to Favorites
Objective	Verify that a publication can be added to the bookmark list successfully.
Preconditions	User has logged into the system.
Test Data	Publication: "Deep Learning in Healthcare"
Test Steps	1. Search for publication "Deep Learning in Healthcare". 2. Open the publication details page. 3. Click the Bookmark button.
Expected Result	The publication is added to the bookmark table successfully.
Actual Result	Bookmark added successfully.
Status	Pass

Field	Description
Module	Bookmark (Favorites)
Test Case ID	TC_BM_02
Test Case Name	Add Duplicate Bookmark
Objective	Verify that duplicate bookmarks are not created.
Preconditions	The publication has already been bookmarked.
Test Data	Publication: “Deep Learning in Healthcare”
Test Steps	1. Open the details page of a bookmarked publication. 2. Click the Bookmark button again.
Expected Result	System displays “This publication is already in your favorites” and no duplicate record is created.
Actual Result	Message displayed correctly and no duplicate record created.
Status	Pass

Field	Description
Module	Bookmark (Favorites)
Test Case ID	TC_BM_03
Test Case Name	Remove Bookmark
Objective	Verify the bookmark removal functionality.
Preconditions	At least one publication exists in the bookmark list.
Test Data	Bookmark ID: 2
Test Steps	1. Open the Favorites page. 2. Select the publication with Bookmark ID = 2. 3. Click the Remove button.
Expected Result	The publication is removed from the bookmark list.
Actual Result	Bookmark removed successfully.
Status	Pass

4.5 View Trending Topics

Field	Description
Module	Topic Following
Test Case ID	TC_TOPIC_01
Test Case Name	Follow New Topic
Objective	Verify that users can successfully follow a new research topic.
Preconditions	User has logged into the system.
Test Data	Topic: Artificial Intelligence
Test Steps	1. Open the Trend Topics page. 2. Select topic "Artificial Intelligence". 3. Click the Follow button.
Expected Result	The topic is stored in the UserFollowTopic table and a new notification is created.
Actual Result	Topic followed successfully and notification created.
Status	Pass

Field	Description
Module	Topic Following
Test Case ID	TC_TOPIC_02
Test Case Name	Follow Existing Topic
Objective	Verify that duplicate topic subscriptions are not created.
Preconditions	The topic has already been followed previously.
Test Data	Topic: Artificial Intelligence
Test Steps	1. Select topic "Artificial Intelligence". 2. Click the Follow button again.
Expected Result	System displays "You are already following this topic" and no new record is created.
Actual Result	Correct message displayed and no duplicate record created.
Status	Pass

Field	Description
Module	Topic Following
Test Case ID	TC_TOPIC_03
Test Case Name	Unfollow Topic
Objective	Verify that users can successfully unfollow a topic.
Preconditions	The topic is currently being followed by the user.
Test Data	Topic: Artificial Intelligence
Test Steps	1. Open the Followed Topics list. 2. Select topic "Artificial Intelligence". 3. Click the Unfollow button.
Expected Result	The topic is removed from the UserFollowTopic table.
Actual Result	Topic unfollowed successfully.
Status	Pass

4.6 Notification Management

Field	Description
Module	Notification Management
Test Case ID	TC_NOTI_01
Test Case Name	View Notification List
Objective	Verify that users can view all notifications associated with their account.
Preconditions	User has logged into the system and has at least one notification.
Test Data	User ID: 1
Test Steps	1. Log in to the system. 2. Click the Notification icon. 3. Open the Notification page.
Expected Result	The system displays all notifications belonging to the user, including title, content, and status.
Actual Result	Notification list displayed successfully.
Status	Pass

Field	Description
Module	Notification Management
Test Case ID	TC_NOTI_02
Test Case Name	Mark Notification as Read
Objective	Verify that users can mark a notification as read.
Preconditions	User has at least one unread notification.
Test Data	Notification ID: 5
Test Steps	1. Open the Notification page. 2. Select an unread notification. 3. Click the notification to view details.
Expected Result	The notification status changes from “Unread” to “Read”.
Actual Result	Notification status updated successfully.
Status	Pass

4.7 Trend Analysis

Field	Description
Module	Trend Analysis
Test Case ID	TC_TREND_01
Test Case Name	Successful Trend Analysis
Objective	Verify that the system correctly generates publication trend analysis for a valid research topic based on synchronized publication metadata stored in the database.
Preconditions	Publication metadata for the topic "Machine Learning" has been synchronized into the database from Semantic Scholar, OpenAlex, and Crossref.
Test Data	Topic: Machine Learning Year From: 2020 Year To: 2024
Test Steps	1. Log in to the system. 2. Access "Track Publication Trends". 3. Enter topic "Machine Learning". 4. Select year range from 2020 to 2024. 5. Click Analyze.
Expected Result	The system aggregates publication counts by year from the database, calculates growth rate, and displays trend statistics and visual charts showing increasing or decreasing publication activity.
Actual Result	Trend analysis results displayed correctly.
Status	Pass

Field	Description
Module	Trend Analysis
Test Case ID	TC_TREND_02
Test Case Name	Trend Analysis with No Data
Objective	Verify system behavior when no trend data is available.
Preconditions	System is available and database contains no matching publication data for the topic
Test Data	Topic: abcdeftopic
Test Steps	1. Enter non-existing topic "abcdeftopic". 2. Click Analyze.
Expected Result	An empty result list is displayed and no system error occurs.
Actual Result	Empty result list displayed correctly.
Status	Pass